

ENGLISH FOR HIGHER EDUCATION AND RESEARCH

Higher Education & Research English

Define the question. Defend the evidence.

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Eight lessons. Eleven extended conversations.

INSIDE THE BOOK

192 vocabulary entries / 128 reusable phrases
220 extended-dialogue turns / 11 transfer exchanges
Word banks and explained answer keys

Upper-intermediate to advanced (B2-C1). For researchers, doctoral students, research coordinators, and academic collaborators.

2026 edition

Eight scenarios. Eight skills.

Eight research conversations: sharpen a study question, bound a grant aim, investigate a lab discrepancy, clarify data reuse, agree credit, respond to review, document a dataset, and answer a conference challenge.

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Answer the challenge without claiming universal proof | Give a concise evidence-based conference answer, distinguish task performance from mechanism, and handle an unresolved question.

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Strong claims need clear boundaries

Academic confidence comes from showing exactly what a study can support. Connect the question to the design, explain the evidence, and distinguish a promising next step from a result already established.

Define what the study measures

Population, outcome, comparison, and timing change the meaning of a question. A broad word such as works can conceal several different claims.

"We measured course completion in this cohort, not the program's causal effect on all students."

Make disagreement inspectable

Point to the protocol, data, criterion, or manuscript passage at issue. A direct challenge can be a request for evidence rather than a personal attack.

"The exposure time changed between runs; we need to check its contribution to the discrepancy."

Keep permission separate from possibility

Having data, software access, or a technically feasible plan does not settle whether the proposed use is authorized.

"We can describe the proposed analysis, but the required reuse determination is still pending."

Respond at the right scale

Answer the narrow question first, then state the limitation and next test. Avoid turning a pilot, one sample, or a successful rerun into universal proof.

"This result supports the reported observation in our sample; broader testing is still needed."

PROFESSIONAL BOUNDARIES

All cases and figures are fictional. This book teaches research communication, not statistical, clinical, or legal advice or ethics approval. Follow the relevant institution, funder, jurisdiction, and journal requirements; obtain required review before conducting or changing research.

What exactly would it mean for tutoring to work?

A Read the situation

Specify a research question and distinguish an observed association from a causal effect.

Doctoral researcher Imani and supervisor Leon revise a proposal asking whether optional tutoring works. Available records cover first-year students in one introductory biology course for one semester: 32 of 40 tutoring participants completed the course, compared with 36 of 60 nonparticipants. Tutoring was self-selected, not randomly assigned. The groups may differ in prior preparation or motivation, which these records do not measure. The immediate analysis can describe the association with completion in this cohort. A future causal study would need a suitable design; it has not been conducted.

PEOPLE / Imani: Doctoral researcher | Leon: Research supervisor

NARROWING A CLAIM STRENGTHENS THE QUESTION

A supervisor who challenges the word works may be asking for a testable question, not rejecting the project. Respond with the population, measured outcome, and comparison. Keep the broader ambition visible as a future question instead of presenting it as an answer from the current records.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What is the completion rate among tutoring participants?

- | | |
|--------------|---------------|
| A 80% | B 60% |
| C 32% | D 125% |

2. What is the difference between the two completion rates?

- | | |
|----------------------------------|-----------------------------------|
| A Forty percentage points | B Twenty percentage points |
| C Twenty percent relative | D Eight percentage points |

3. Why is a causal claim unsupported here?

- | | |
|---|--|
| A Both groups have identical completion rates. | B Tutoring was self-selected and relevant group differences are unmeasured. |
| C Every observational record is useless. | D The course has no defined outcome. |

Answers and explanations: [page 85](#).

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

research question

A specific question a study is designed to answer.

IN USE refine a research question

target population

The group to which a research question refers.

IN USE define the target population

study sample

The observed group included in the study.

IN USE describe the study sample

cohort

A defined group observed over a specified period.

IN USE follow a student cohort

outcome variable

The measured result of interest.

IN USE specify the outcome variable

exposure

A condition or experience whose relationship with an outcome is studied.

IN USE define the exposure

comparison group

The group used as a reference for a contrast.

IN USE identify the comparison group

observational study

A study observing conditions without assigning the exposure.

IN USE describe an observational study

self-selection

Entry into a group through participants' own choices.

IN USE account for self-selection

random assignment

Allocation to study conditions using a random process.

IN USE distinguish random assignment

random sampling

Selection of units from a population by a random process.

IN USE distinguish random sampling

association

An observed relationship between variables.

IN USE report an association

Precision check

32/40 is 80%; 36/60 is 60%. The difference is 20 percentage points. Relative to 60%, the difference is approximately 33.3%, not 20%. Neither calculation removes the design limitation created by self-selection.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

causal effect

A change attributable to an intervention or exposure.

IN USE estimate a causal effect

confounder

A factor influencing both exposure and outcome that can distort their apparent causal relationship.

IN USE address potential confounders

baseline difference

A group difference present before the studied exposure or outcome.

IN USE examine baseline differences

operational definition

A precise specification of how a concept is measured.

IN USE state an operational definition

completion rate

The proportion meeting the defined completion outcome.

IN USE calculate the completion rate

denominator

The base count used in a proportion.

IN USE state the denominator

percentage-point difference

The arithmetic difference between percentages.

IN USE report a percentage-point difference

relative difference

A difference expressed in relation to a reference value.

IN USE distinguish a relative difference

internal validity

The credibility of the study's inference within its setting.

IN USE assess internal validity

external validity

The extent to which findings apply beyond the studied setting.

IN USE discuss external validity

estimand

The precise quantity a study seeks to estimate.

IN USE define the estimand

study protocol

A documented plan for the study's conduct and analysis.

IN USE revise the study protocol

Precision check

Random assignment concerns allocation to conditions; random sampling concerns selection into the sample. They address different questions. The current case has self-selected tutoring participation, and no broader representativeness or causal design is established.

Phrases you can actually use

C Notice the language / 1 of 2

CLARIFY THE BROAD VERB

"What outcome would count as working?"

SET THE PERIOD

"The observation window is one semester."

IDENTIFY EXPOSURE

"Tutoring participation was optional."

STATE THE COMPARISON

"Thirty-six of sixty nonparticipants completed, or 60%."

SPECIFY THE POPULATION

"The records cover first-year students in this course."

NAME THE OUTCOME

"We are measuring course completion."

STATE THE FIRST RATE

"Thirty-two of forty participants completed, or 80%."

EXPRESS THE CONTRAST

"That is a 20-percentage-point difference."

Works

Needs a specified outcome, population, comparison, and time.

Completed

Requires a consistent completion definition for both groups.

Higher

Describes direction without establishing cause.

Phrases you can actually use

C Notice the language / 2 of 2

QUALIFY THE INFERENCE

"This is an observed association, not an isolated causal effect."

IDENTIFY MISSING INFORMATION

"Prior preparation and motivation are not measured here."

BOUND THE POPULATION CLAIM

"The current result concerns this course cohort."

PRESERVE THE USEFUL RESULT

"The descriptive contrast remains worth reporting."

NAME SELECTION

"Students chose whether to participate."

AVOID INVENTING AN EXPLANATION

"Those are possible group differences, not confirmed causes."

SEPARATE FUTURE WORK

"A causal question needs a suitable future design."

CLOSE THE REVISION

"The proposal should match its question to the available evidence."

Associated with

Does not mean caused by.

Random

State whether it concerns sampling or assignment.

May differ

Marks a possible explanation, not a finding.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which headline fits the available evidence?

- A** Tutoring participation was associated with higher completion in this course cohort.
- B** Tutoring increased completion by twenty percentage points in this course cohort.
- C** Tutoring participants had twenty percent higher completion than nonparticipants.
- D** Randomly selecting more records would establish the tutoring effect.

2. Which question is still unresolved?

- A** Whether the course records span one semester
- B** Whether participation was optional
- C** Whether tutoring caused the completion difference
- D** Whether 32 of 40 equals 80%

3. Which statement about possible confounding is accurate?

- A** Motivation is proved to explain the entire contrast.
- B** Self-selection makes confounding impossible.
- C** A percentage-point calculation adjusts for preparation.
- D** Prior preparation or motivation may differ, but these records do not establish those differences.

4. Which design distinction is correct?

- A** Random assignment guarantees every population is represented.
- B** Random sampling automatically makes an optional program randomized.
- C** Both terms mean the same operation.
- D** Random assignment allocates conditions; random sampling selects sample members.

Answers and explanations: [page 85](#).

What exactly would it mean for tutoring to work?

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

outcome variable / denominator / causal effect / Random assignment / self-selection / study protocol / research question / external validity / percentage-point difference / association

- 01 **Leon:** The proposal asks whether tutoring works. That sounds important, but what outcome would answer the question, and whose experience are we actually observing?
- 02 **Imani:** I need a sharper ¹ _____. The available records concern first-year students in one introductory biology course during one semester. They record tutoring participation and course completion.
- 03 **Leon:** Then completion, rather than examination performance or long-term retention, is the outcome. We should not let the proposal switch among those meanings without noticing.
- 04 **Imani:** Agreed. The ² _____ is completion as recorded for that semester. I will use the same definition for participants and nonparticipants, so the comparison does not change meaning between groups.
- 05 **Leon:** Walk me through the counts before we discuss the interpretation. Percentages can conceal different group sizes, especially when the audience sees only a headline.
- 06 **Imani:** The participant ³ _____ is forty, with thirty-two completing. That is eighty percent. Among sixty nonparticipants, thirty-six completed, giving sixty percent for the comparison group.
- 07 **Leon:** The draft says twenty percent higher. Do you mean twenty percentage points, or are you calculating a relative increase from the nonparticipant rate?
- 08 **Imani:** Correct. The ⁴ _____ is twenty. Relative to sixty percent, the difference is about thirty-three point three percent. Neither phrasing, by itself, explains why the groups differ.
- 09 **Leon:** Were students allocated to tutoring, or did they choose it? That matters more for the causal claim than how confidently we present the percentages.
- 10 **Imani:** Participation involved ⁵ _____. Students chose whether to attend. The records do not measure prior preparation or motivation, and either could differ between groups. We have not established those differences.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 10.

- 11 **Leon:** Then the claim that tutoring caused the higher completion rate is stronger than the design supports. We can still report the observed relationship accurately.
- 12 **Imani:** I will call it an ⁶ _____ with completion in this cohort. That preserves the finding without treating possible unmeasured group differences as either irrelevant or already proved explanations.
- 13 **Leon:** A colleague proposed solving the problem by randomly sampling more records from the same optional program. Would that make tutoring itself randomly assigned?
- 14 **Imani:** No. ⁷ _____ concerns allocation to conditions. Random sampling concerns which people enter the sample. A larger randomly sampled record set would not retroactively randomize the tutoring decision.
- 15 **Leon:** Good. We should also avoid implying that one course at one institution establishes what happens for all students in every subject and setting.
- 16 **Imani:** That is an ⁸ _____ question. The current description stays with this course and semester. Broader applicability needs evidence across the relevant settings, not just a less cautious title.
- 17 **Leon:** Let us keep the causal question for the next study. For this paper, can you put the course, semester, and two counts into the abstract?
- 18 **Imani:** The future ⁹ _____ question needs a suitable design and explicit assumptions. We should discuss that design as a proposed next step, not imply that it has already been implemented.
- 19 **Leon:** Revise the objective to describe completion by tutoring participation in this cohort. Then make the design limitation visible in the abstract as well.
- 20 **Imani:** I will update the ¹⁰ _____ and the proposal wording together. The useful result is a bounded comparison with clear counts, while the causal and broader-population questions remain open.

ANSWERS / EXPLANATIONS: PAGE 86

1. research question | 2. outcome variable | 3. denominator | 4. percentage-point difference | 5. self-selection | 6. association | 7. Random assignment | 8. external validity | 9. causal effect | 10. study protocol

A rate difference is not a randomized effect

F Say it

In a self-selected workshop cohort, 18 of 30 attendees and 20 of 40 nonattendees completed a task. Prior experience was not measured. The records cover one course only.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

causal effect / 50 / 60 / 10

Researcher: "The attendee completion rate is ¹ _____ percent."

Supervisor: "The nonattendee rate is ² _____ percent."

Researcher: "The difference is ³ _____ percentage points."

Supervisor: "The contrast establishes an association, not a ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 86

1. 60 | 2. 50 | 3. 10 | 4. causal effect

Rehearse with a fixed script

1. Read turns 5-8, stressing each denominator and the words percentage points.
2. Repeat turns 9-14, distinguishing self-selection, random sampling, and random assignment.
3. Switch roles for the transfer and read the three completed calculations with their explanations.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

A feasibility pilot cannot promise a definitive answer

A Read the situation

Align a grant aim with pilot scope and interpret predefined feasibility milestones.

Research lead Sora and grants adviser Ben revise a proposal for a small learning-study pilot. The draft promises to establish that a new workshop improves academic attainment. The actual pilot tests recruitment and follow-up procedures, not efficacy. A completed preparatory pilot recruited 30 students in four weeks and retained 24 at follow-up. The team had specified two local progression criteria: recruit at least 30 in four weeks and retain at least 85% of those recruited. Both must be met for automatic progression to the next planning stage. Only recruitment passed; a redesign or explicit review is needed.

PEOPLE / Sora: Research lead | Ben: Grants adviser

AMBITION AND DELIVERABLES ARE DIFFERENT

A proposal can explain a significant long-term problem without promising that a small pilot will solve it. State the pilot deliverable precisely and show how it informs a later study. Credible limits help reviewers understand what their funding would actually produce.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What is the pilot designed to assess?

A All future funding decisions

B Recruitment and follow-up feasibility

C Definitive improvement in attainment

D Universal workshop safety

2. What was the observed retention rate?

A 125%

B 80%

C 85%

D 24%

3. Did both local progression criteria pass?

A No; recruitment passed and retention did not.

B Yes, because thirty students were recruited.

C No; retention passed and recruitment did not.

D Yes, because eighty and eighty-five are equivalent.

Answers and explanations: page 87.

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

specific aim

A focused objective a proposed study will address.

IN USE refine a specific aim

long-term goal

The broader result a research program ultimately seeks.

IN USE state the long-term goal

pilot study

A small-scale study testing procedures for later work.

IN USE design a feasibility pilot

feasibility

Whether proposed methods can be carried out practicably.

IN USE assess feasibility

efficacy

The effect of an intervention under the relevant controlled conditions.

IN USE distinguish efficacy from feasibility

recruitment

Bringing eligible participants into a study.

IN USE monitor recruitment

recruitment window

The period allowed for enrolling participants.

IN USE define the recruitment window

retention

Continued participation through a defined study point.

IN USE calculate follow-up retention

follow-up

A later observation or contact after the initial stage.

IN USE complete follow-up

attrition

Loss of participants from later study observation.

IN USE report participant attrition

progression criterion

A predefined condition for moving to a next stage.

IN USE apply progression criteria

milestone

A specified achievement used to assess progress.

IN USE set a measurable milestone

Precision check

24/30 is 80% retention, below the local 85% criterion. With 30 recruited participants, at least 26 retained would reach or exceed 85%; 25 would be only about 83.3%. Do not round a failed threshold into a pass.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

deliverable

A defined output the project will provide.

IN USE identify the pilot deliverable

study objective

The particular purpose of a study.

IN USE align the study objective

rationale

The reasoning that justifies the proposed work.

IN USE strengthen the study rationale

significance

Why the research question or gap matters.

IN USE explain research significance

innovation

A meaningful new feature of an approach.

IN USE describe the innovation precisely

approach

The planned methods for addressing the aims.

IN USE evaluate the proposed approach

scope

The boundaries of the work and claims.

IN USE keep the aims within scope

sample size

The number of units or participants included.

IN USE justify the sample size

precision

The degree of uncertainty around an estimate.

IN USE discuss estimate precision

definitive study

A study designed to provide a robust answer to the main question.

IN USE plan a definitive study

contingency

A planned response if assumptions or milestones fail.

IN USE specify a contingency

funding decision

A determination about financial support.

IN USE distinguish a funding decision

Precision check

The two criteria are conjunctive: both are required for automatic progression. Recruitment success does not cancel the retention shortfall. Failing this local rule calls for redesign or review; it does not prove the workshop has no educational effect.

Phrases you can actually use

C Notice the language / 1 of 2

SEPARATE AMBITION FROM AIM

"The long-term goal is broader than this pilot deliverable."

REMOVE AN OVERCLAIM

"It is not designed to establish efficacy."

REPORT RECRUITMENT

"That target was met."

GIVE THE RESULT

"Twenty-four were retained, which is 80%."

STATE THE PILOT PURPOSE

"This study tests recruitment and follow-up procedures."

NAME THE RECRUITMENT TARGET

"The local target was 30 students in four weeks."

STATE THE RETENTION BASE

"Retention is calculated among the 30 recruited students."

APPLY THE THRESHOLD

"That is below the predefined 85% criterion."

Pilot

State whether it tests feasibility rather than definitive effects.

At least

Includes the threshold and values above it.

Retained

Needs a specified follow-up point and denominator.

Phrases you can actually use

C Notice the language / 2 of 2

KEEP BOTH CONDITIONS

"Automatic progression requires both criteria."

EXPLAIN THE NEXT STEP

"The shortfall needs redesign or explicit review."

SPECIFY THE OUTPUT

"The deliverable is a revised, tested study procedure."

BOUND FUNDING LANGUAGE

"Meeting a local milestone would not guarantee an award."

AVOID CONVENIENT ROUNDING

"Eighty percent cannot be reported as meeting 85%."

PRESERVE UNCERTAINTY

"This does not establish that the workshop is ineffective."

RETAIN SIGNIFICANCE

"The larger educational question remains important."

CLOSE THE AIM

"The proposal should promise what this design can deliver."

Both

One successful criterion is insufficient.

Automatic progression

A local planning rule, not a funding guarantee.

Did not meet

Does not establish that the intervention has no effect.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which specific aim matches the supplied pilot?

- A** Demonstrate that the observed retention rate met both predefined criteria.
- B** Assess recruitment and follow-up procedures to refine a later study.
- C** Estimate workshop efficacy by treating retained participants as improved learners.
- D** Establish feasibility from recruitment alone, leaving follow-up outside the progression rule.

2. How many retained participants would meet at least 85% of thirty?

- A** Twenty-four
- B** Twenty-five
- C** Twenty
- D** Twenty-six

3. What is the correct progression statement?

- A** Retention should be rounded up to the target.
- B** The target can be changed silently after seeing the result.
- C** Recruitment passed, but retention requires redesign or explicit review before progression.
- D** One passed milestone automatically overrides the other.

4. What does the retention result say about efficacy?

- A** It proves a large effect.
- B** It directly measures examination improvement.
- C** It does not establish whether the workshop improves attainment.
- D** It proves the workshop never works.

Answers and explanations: page 87.

A feasibility pilot cannot promise a definitive answer

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

Feasibility / progression criterion / retention / contingency / recruitment / deliverable / rationale / scope / funding decision / specific aim

- 01 **Ben:** The draft says this pilot will establish that the workshop improves attainment. Is that really the question the small study was designed to answer?
- 02 **Sora:** No. The ¹ _____ should concern recruitment and follow-up procedures. Improving attainment is our long-term goal, but the pilot does not provide a definitive test of that outcome.
- 03 **Ben:** We can keep the wider educational problem in the significance section. Reviewers still need a concrete account of what this grant would deliver now.
- 04 **Sora:** The ² _____ is a refined study procedure, informed by recruitment and follow-up evidence. That is useful preparation for a later study, without pretending the larger question has already been answered.
- 05 **Ben:** The preparatory work enrolled thirty students in four weeks. Remind me whether that matches the target set before the results were available.
- 06 **Sora:** It meets the ³ _____ criterion of at least thirty within four weeks. We should report that success, while checking the other criterion separately instead of treating enrollment as the whole feasibility assessment.
- 07 **Ben:** Can we replace good retention with the actual rate? We kept twenty-four of thirty, and the milestone table still says eighty-five percent.
- 08 **Sora:** The ⁴ _____ rate is eighty percent. We had set at least eighty-five percent, so this criterion was not met. The denominator is all thirty recruited students.
- 09 **Ben:** Would twenty-five retained students meet the threshold? We need to translate the percentage into a whole-participant count.
- 10 **Sora:** The ⁵ _____ requires at least twenty-six retained out of thirty. Twenty-five would be approximately eighty-three point three percent, still below the threshold. We should apply the stated rule accurately.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 19.

- 11 **Ben:** Then the milestone table needs one pass and one shortfall. I cannot sign off automatic progression on the recruitment number alone.
- 12 **Sora:** We need a ⁶ _____ such as redesign or explicit review. We should explain what the shortfall means for planning, not silently lower the criterion because we prefer a positive result.
- 13 **Ben:** That does not mean the workshop is educationally ineffective. The missing follow-up information concerns the study procedure, and the pilot was not an efficacy trial.
- 14 **Sora:** Correct. ⁷ _____ and efficacy are different questions. The result tells us the retention procedure missed our local target; it does not establish whether the workshop improves attainment.
- 15 **Ben:** I will also remove the sentence promising a future award once the pilot milestones pass. Our internal progression rule cannot bind a funder's decision.
- 16 **Sora:** A ⁸ _____ remains separate. Even successful local milestones would inform a proposal, not guarantee financial support. We should state the value of the evidence without predicting an external decision.
- 17 **Ben:** The grant can remain ambitious: a clear problem, a bounded pilot, and a reasoned route toward a definitive study.
- 18 **Sora:** Yes. Keep the ⁹ _____ explicit. We can describe the future question and explain how improving follow-up supports it, while keeping the current work focused on procedures and practical uncertainty.
- 19 **Ben:** I will revise the aims, summary, and milestone table together. They should all say recruitment passed, retention did not, and the next planning step requires review.
- 20 **Sora:** That gives us a consistent ¹⁰ _____ for the proposed work. We are asking for support to resolve a concrete procedural problem, not claiming that a small pilot has settled the educational effect.

ANSWERS / EXPLANATIONS: PAGE 88

1. specific aim | 2. deliverable | 3. recruitment | 4. retention | 5. progression criterion | 6. contingency | 7. Feasibility | 8. funding decision | 9. scope | 10. rationale

Two conditions, not one

F Say it

A local pilot rule requires 20 recruits within three weeks and at least 90% retention. Twenty are recruited on time, but only 17 reach follow-up. Both conditions are needed for automatic progression.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

met / not supported / 85 / 18

Researcher: "The recruitment condition was ¹ _____."

Adviser: "Retention was ² _____ percent."

Researcher: "At least ³ _____ retained students were required."

Adviser: "Automatic progression is ⁴ _____ under the stated rule."

ANSWERS / EXPLANATIONS: PAGE 88

1. met | 2. 85 | 3. 18 | 4. not supported

Rehearse with a fixed script

1. Read turns 5-12, stating 24 out of 30 and the minimum qualifying count of 26 clearly.
2. Repeat turns 13-16, separating feasibility, efficacy, and the funding decision.
3. Switch roles for the transfer; read the 85% observation and 90% requirement without rounding one into the other.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

The repeated run was not performed the same way

03

A Read the situation

Present an experimental discrepancy, disclose a method change, and propose a controlled check without editing the record to fit a preferred result.

Research associate Alex and lab lead Priya compare two nonclinical fluorescence measurements of the same reference material. Run A reports 120 arbitrary units and Run B 180. Instrument files reveal exposure times of ten seconds for A and twenty for B. The summary lab notes had omitted this change. The team has not tested whether it explains the entire difference, and signal response may not be proportional to exposure. They must retain both raw records, add a dated correction to the notes, and plan a repeat with documented matched settings. No misconduct finding has been made.

PEOPLE / Alex: Research associate | Priya: Lab lead

DISCLOSE A DISCREPANCY WITHOUT ASSIGNING BLAME

A lab meeting should make uncertainty inspectable. Name the observed result, the method difference, and the missing evidence. Correct documentation transparently without accusing a colleague of misconduct or presenting an untested explanation as the established cause.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What changed between the runs?

- A** The notes document identical exposure times.
- B** Run A used no reference material.
- C** Run B's result was 120 units.
- D** Exposure time increased from ten to twenty seconds.

2. What is not yet established?

- A** Whether a method change was omitted
- B** Whether raw records exist
- C** Whether exposure time explains the entire signal difference
- D** Whether the reported values are 120 and 180

3. How should the documentation be corrected?

- A** Backdate a replacement note as if nothing changed.
- B** Delete the less convenient raw result.
- C** Announce proven misconduct immediately.
- D** Add a dated correction while preserving the original record.

Answers and explanations: [page 89](#).

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

lab meeting

A discussion of research progress, methods, and results.

IN USE present at a lab meeting

experimental run

A recorded execution of a measurement or experiment.

IN USE compare experimental runs

reference material

Material used as a stable point of comparison.

IN USE measure the reference material

raw data

Original recorded observations before later processing.

IN USE preserve raw data

processed result

A value produced after defined analysis or transformation.

IN USE trace a processed result

arbitrary unit

A relative measurement unit defined by the measurement system.

IN USE report arbitrary units

exposure time

The duration used for signal acquisition in this case.

IN USE document exposure time

instrument setting

A parameter controlling measurement behavior.

IN USE match instrument settings

protocol version

The identified revision of a study procedure.

IN USE record the protocol version

method deviation

A departure from the intended or comparison procedure.

IN USE document a method deviation

batch effect

A systematic difference associated with processing groups.

IN USE investigate a batch effect

technical replicate

A repeated measurement used to assess measurement variability.

IN USE compare technical replicates

Precision check

Run B is 60 arbitrary units higher, a 50% increase relative to Run A. Exposure doubled, but the signal increased by only half. Do not divide or scale signals as though proportional response had been established.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

biological replicate

An independently derived biological sample or unit.

IN USE distinguish biological replicates

signal intensity

The magnitude of the detected measurement signal.

IN USE compare signal intensity

linear response

A straight-line input-output relationship over a specified range, possibly with a nonzero offset.

IN USE verify a linear response

saturation

A measurement ceiling beyond which more input is not represented proportionally.

IN USE check for signal saturation

normalization

Adjustment to a defined reference or scale.

IN USE justify normalization

measurement variability

Variation arising across repeated measurements.

IN USE quantify measurement variability

controlled comparison

A comparison with relevant conditions deliberately held consistent.

IN USE plan a controlled comparison

lab notebook

The record of research methods, observations, and changes.

IN USE maintain the lab notebook

dated correction

A visible amendment recording when a correction was made.

IN USE add a dated correction

provenance

The origin and history of data or records.

IN USE preserve data provenance

hypothesis

A proposed explanation or prediction to be tested.

IN USE test a methodological hypothesis

repeatability check

A check of measurement agreement under the same specified short-term conditions.

IN USE document a repeatability check

Precision check

A recorded method difference is a candidate explanation, not proof of the entire cause. Retain both results and document the correction. Repeating measurements of the same material does not create new independent biological samples.

Phrases you can actually use

C Notice the language / 1 of 2

03

OPEN WITH THE RESULT

"Run A gave 120 units; Run B gave 180."

STATE THE DURATIONS

"A used ten seconds and B used twenty."

PRESERVE THE RECORDS

"Keep both raw files and their original settings."

AVOID BLAME

"This is a documentation gap, not a misconduct finding."

NAME THE VERIFIED CHANGE

"The instrument files show different exposure times."

DISCLOSE THE OMISSION

"The summary notes did not record that change."

CORRECT TRANSPARENTLY

"Add a dated correction to the notebook."

LIMIT THE EXPLANATION

"The change may contribute, but we have not isolated its effect."

Repeated

Does not guarantee identical conditions.

Raw

Keep the original record separate from later transformations.

May explain

A hypothesis, not a completed causal test.

Phrases you can actually use

C Notice the language / 2 of 2

03

QUESTION A SHORTCUT

"Proportional scaling has not been validated."

PLAN THE COMPARISON

"Repeat with documented matched settings."

TRACK PROCESSING

"State every transformation applied to the raw values."

MAKE THE ACTION TRACEABLE

"Link the repeat to the protocol version and instrument files."

CHECK THE RESPONSE

"We need to examine linearity and possible saturation."

KEEP REPLICATION PRECISE

"These are repeated measurements of the same material."

SEPARATE OBSERVATION AND CAUSE

"The signal changed; the complete cause is unresolved."

CLOSE HONESTLY

"We have a testable explanation, not a resolved discrepancy."

Twice the exposure

Does not establish twice the signal.

Correction

Must remain distinguishable from the original entry.

Independent sample

Not created by measuring the same material again.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which sentence accurately reports the signal contrast?

- A** Run A is 180 units.
- B** The signal difference is exactly zero.
- C** Run B is 60 units, or 50%, higher than Run A.
- D** Run B doubled from 120 to 240.

2. Why is dividing Run B by two not justified as a correction?

- A** The exposure times are identical.
- B** Arbitrary units mean the records have no information.
- C** Proportional signal response to exposure has not been established.
- D** Division is never used in research.

3. What is the next useful comparison?

- A** Selecting whichever result supports the preferred theory
- B** A documented repeat with matched settings and preserved raw records
- C** A repeat with undisclosed settings
- D** Deleting Run B

4. Which conclusion should be avoided?

- A** Exposure differs between runs.
- B** The complete cause remains unresolved.
- C** The omitted note proves research misconduct.
- D** The summary record needs correction.

Answers and explanations: [page 89](#).

The repeated run was not performed the same way

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

linear response / dated correction / provenance / instrument setting / controlled comparison / saturation / raw data / repeatability check / hypothesis / technical replicates

- 01 **Priya:** The slide calls Run B a repeat, but its signal is one hundred eighty units rather than one hundred twenty. Were the measurement conditions actually matched?
- 02 **Alex:** The ¹ _____ for exposure changed. The files show ten seconds for Run A and twenty for Run B. That difference was omitted from the summary notes.
- 03 **Priya:** Preserve both files, including the unexpected result, before correcting the notes. We need the original evidence intact.
- 04 **Alex:** I have retained the ² _____ and the original metadata. I will not replace either measurement with an adjusted value. Any later processing must remain distinguishable from what the instrument originally recorded.
- 05 **Priya:** The exposure doubled, so somebody suggested halving the second signal and treating that as the comparable result. Have we established a proportional response?
- 06 **Alex:** No. A ³ _____ has not been verified here. Even a straight-line response could include background signal. We cannot simply halve one raw value and call it corrected.
- 07 **Priya:** Check whether the detector reaches a ceiling or other conditions differed. The exposure change is a lead, not a complete explanation.
- 08 **Alex:** We can check for ⁴ _____ and review other recorded conditions. I will describe these as possible contributors, not confirmed causes. We have not isolated the exposure effect or explained the full discrepancy.
- 09 **Priya:** How will the correction show when we discovered the omitted setting, without hiding the original note?
- 10 **Alex:** I will add a ⁵ _____ linked to the instrument files. It will state the actual exposure for each run and when the omission was identified, without backdating the new entry.

The conversation, continued

E Conversations / turns 11-20

03

Continue with the word bank on page 28.

- 11 **Priya:** Good. That also avoids implying misconduct before anything has been investigated. The immediate issue is an incomplete method record and a result needing explanation.
- 12 **Alex:** Preserving ⁶ _____ keeps the measurement, original notes, and later correction connected. I will record what changed and when, without assigning blame before the facts are checked.
- 13 **Priya:** For the next measurement, we need documented matched settings and a clear plan. Simply running the instrument again would not address the comparison problem.
- 14 **Alex:** I will propose a ⁷ _____ using matched settings and retained raw files. We should specify the protocol version and checks in advance, so another undocumented change does not create the same uncertainty.
- 15 **Priya:** For that repeat, keep the operator, instrument, location, and procedure the same within one session. We are testing measurement consistency, not collecting new biological samples.
- 16 **Alex:** Those planned measurements are ⁸ _____. The original exposure settings differed, so those two runs are not a matched set. Neither repeating the measurement nor changing exposure adds an independent biological sample.
- 17 **Priya:** The slide should report the observed difference as sixty units, fifty percent relative to Run A. It must not say exposure caused exactly that increase.
- 18 **Alex:** I will separate the observation from the ⁹ _____. The documented change is a plausible contributor. Its role remains to be tested, and other sources of variation have not been ruled out.
- 19 **Priya:** Bring both raw files, the corrected record, and the matched-settings plan next time. We can then review the evidence together.
- 20 **Alex:** I will label the planned work a ¹⁰ _____. The two original runs and their settings stay in the record. We still need to explain the discrepancy; scheduling a repeat does not close it.

ANSWERS / EXPLANATIONS: PAGE 90

1. instrument setting | 2. raw data | 3. linear response | 4. saturation | 5. dated correction | 6. provenance | 7. controlled comparison | 8. technical replicates | 9. hypothesis | 10. repeatability check

Preserve the change history

F Say it

03

Two readings of the same reference material used different documented gain settings. The summary omitted the change. No causal check has been completed. Both original files remain available.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

untested / difference / dated correction / preserved

Researcher: "The gain change is a verified method ¹ _____."

Lead: "Its causal contribution remains ² _____."

Researcher: "The notebook needs a ³ _____."

Lead: "Both original files should be ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 90

1. difference | 2. untested | 3. dated correction | 4. preserved

Rehearse with a fixed script

1. Read turns 1-10; distinguish the recorded setting change from an untested explanation.
2. Repeat turns 13-16, stressing same session and technical replicates.
3. Switch roles for the transfer; use a firm, neutral tone for the correction without implying misconduct.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

Possessing the data is not permission for a new use

04

A Read the situation

Describe a secondary-use proposal and seek an authorized ethics determination without assuming coded data is anonymous.

Researcher Tessa wants to reuse coded interviews from a student-stress study for a new employment-discrimination question. Her research team can access the key linking codes to participants. The original approval and consent materials have not been checked against the new purpose. The fictional institution requires its designated ethics office to determine the permitted route before new analysis or sharing begins. No determination, amendment, waiver, or new consent has been granted. Tessa can prepare the proposal and gather existing documents, but must not begin the proposed reuse or promise collaborators access.

PEOPLE / Tessa: Researcher | Omar: Research ethics coordinator

A REVIEW BOUNDARY IS NOT A SCIENTIFIC REJECTION

An ethics coordinator may support the question while withholding permission for the proposed use. Explain the purpose and access conditions accurately. Do not turn a discussion about a possible route into a claim that approval, exemption, or participant agreement has already been obtained.

Briefing checks

Circle one answer in each item. Use the briefing.

1. Why are the coded interviews not automatically anonymous to this team?

- A** Interviews never contain private information.
- B** The new topic removes the original identities.
- C** The team can access the code-to-participant key.
- D** All coded data are public.

2. Who determines the permitted route under the supplied local rule?

- A** The researcher acting alone
- B** The software vendor
- C** The designated ethics office
- D** Any collaborator who wants the data

3. What can Tessa do now?

- A** Start the new analysis immediately.
- B** Promise unrestricted collaborator access.
- C** Declare a waiver has been granted.
- D** Prepare the proposal and gather the existing approval and consent documents.

Answers and explanations: page 91.

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

secondary use

Using previously collected material for a further purpose.

IN USE propose a secondary use

human-subjects research

Research involving people or qualifying information under applicable rules.

IN USE assess human-subjects research status

ethics review

Examination of a research proposal's ethical acceptability and requirements.

IN USE request ethics review

institutional review board

A U.S. committee reviewing covered human-subjects research.

IN USE consult the institutional review board

research ethics committee

A body reviewing research ethics under its remit.

IN USE seek research ethics committee review

informed consent

A process supporting an informed, voluntary participation decision.

IN USE review informed-consent materials

consent scope

The activities and uses covered by the relevant consent.

IN USE check the consent scope

approval scope

The work permitted by a particular authorization.

IN USE verify the approval scope

protocol amendment

A proposed change to an approved research protocol.

IN USE submit a protocol amendment

waiver

An authorized exception to a specified requirement.

IN USE distinguish a requested waiver

exemption determination

An authorized decision about an applicable exemption category.

IN USE obtain an exemption determination

coded data

Data using codes with a separate identity-linking key.

IN USE describe coded data accurately

Precision check

Coded does not automatically mean anonymous. Here, the team can access a linking key, so the identity connection matters. The new question and access arrangements must be described accurately to the designated reviewer.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

linking key

Information connecting study codes to identities.

IN USE restrict access to the linking key

identifiability

The possibility of connecting information with a person.

IN USE assess identifiability

de-identification

Removing or reducing identifying information under a specified method.

IN USE document de-identification

confidentiality

Protection against unauthorized information disclosure.

IN USE maintain confidentiality

data-use agreement

Terms governing permitted access and use.

IN USE establish a data-use agreement

authorized determination

A decision made by the designated competent authority.

IN USE await an authorized determination

recontact

Contacting a participant again after earlier involvement.

IN USE assess a recontact proposal

participant information

Material explaining a study to potential participants.

IN USE review participant information

voluntariness

Freedom to choose participation without improper pressure.

IN USE protect voluntariness

access restriction

A limit on who may obtain or use information.

IN USE apply access restrictions

purpose change

A shift from the original use to a different objective.

IN USE describe the purpose change

review status

The current stage or outcome of a review request.

IN USE state the review status

Precision check

A request for an amendment, waiver, or exemption is not a granted determination. The applicable route depends on the study and rules. This scenario supplies a local hold on new analysis and sharing until the ethics office determines that route.

Phrases you can actually use

C Notice the language / 1 of 2

DESCRIBE THE NEW QUESTION

"The proposed reuse concerns employment discrimination."

STATE THE KEY ACCESS

"Our team can access the code-to-participant key."

CHECK THE DOCUMENTS

"We need the original approval and consent materials."

NAME THE AUTHORITY

"The designated ethics office determines the permitted route."

IDENTIFY THE ORIGINAL PURPOSE

"The interviews were collected for a student-stress study."

AVOID AN ANONYMITY CLAIM

"Coding does not make these data anonymous to us."

CLARIFY THE SCOPE

"The new purpose has not been assessed against those documents."

STATE THE HOLD

"Do not start the new analysis or sharing yet."

Coded

A linking key may still allow identification.

Anonymous

Do not use without a justified assessment of the actual situation.

Existing approval

May not cover a new question or sharing arrangement.

Phrases you can actually use

C Notice the language / 2 of 2

KEEP REQUESTS CONDITIONAL

"An amendment may be needed; none has been approved."

PREPARE USEFUL MATERIAL

"We can describe the question, fields, recipients, and safeguards."

LIMIT A COLLABORATOR PROMISE

"Access is subject to the authorized determination."

REPORT STATUS PRECISELY

"The proposal is awaiting review, not approved."

AVOID INVENTING A WAIVER

"No waiver or new consent has been granted."

SEPARATE PREPARATION FROM REUSE

"Gathering documents is not approval to analyze the interviews."

PRESERVE CONFIDENTIALITY

"Keep the current access restrictions in place."

CLOSE WITH THE NEXT STEP

"Submit the complete proposal and wait for the required decision."

May require

A possibility, not a determination already made.

Requested

Not granted.

Preparing a proposal

Distinct from conducting the proposed new research.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which description should be sent to the ethics office?
 - A Public information requiring no consideration
 - B Newly consented data for every conceivable purpose
 - C Coded interviews with a linking key accessible to the research team
 - D Anonymous data with no possible identity connection

2. What does the old approval establish about this new purpose?
 - A Automatic approval for all later questions
 - B A granted waiver of every requirement
 - C Permission to promise external access immediately
 - D Nothing conclusive until its scope is checked against the proposal.

3. Which message to a collaborator is accurate?
 - A "Approval is guaranteed because we have the files."
 - B "All participants have already agreed to this new question."
 - C "A requested exemption is the same as a granted one."
 - D "The proposed access awaits the designated ethics determination."

4. Which preparatory step respects the supplied boundary?
 - A Gather documents and describe the proposed use without beginning analysis or sharing.
 - B Run the analysis first and seek retrospective permission.
 - C Remove the key label but keep unrestricted key access.
 - D Email the full interviews to collaborators now.

Answers and explanations: [page 91](#).

Possessing the data is not permission for a new use

E Conversations / turns 1-10

04

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

secondary use / coded data / protocol amendment / consent scope / linking key / data-use agreement / authorized determination / review status / access restriction / purpose change

- 01 **Tessa:** We already have the interview files, and the new employment-discrimination question is promising. Can I begin analyzing them while we check the old paperwork?
- 02 **Omar:** The local ¹ _____ does not permit that yet. The designated ethics office must determine the route before new analysis or sharing. You can prepare the proposal and gather the existing documents.
- 03 **Tessa:** The interviews came from our student-stress study. They use participant codes rather than names, so I initially described them as anonymous in the draft.
- 04 **Omar:** Describe the ² _____ accurately. Does the research team have access to the key connecting the codes with participants? That access changes what the reviewer needs to know about identifiability.
- 05 **Tessa:** Yes, the team can access the key. The draft should state that plainly rather than imply there is no possible connection with the people interviewed.
- 06 **Omar:** Exactly. The ³ _____ must be included in the access description. Coding alone does not establish anonymity to your team, and changing the label would not change the underlying arrangement.
- 07 **Tessa:** I have the original participant information and consent documents, but I have not checked whether the employment question is within what they describe.
- 08 **Omar:** Review the ⁴ _____ alongside the original approval. We should not assume that agreement to the student-stress study covers every later question or sharing arrangement.
- 09 **Tessa:** I will send both research questions side by side. The employment topic is not actually named in the documents I have opened so far.
- 10 **Omar:** Yes, state the ⁵ _____ clearly. The reviewer needs the actual new question, intended information, recipients, and safeguards, not a broad assurance that everything is still generally about students.

The conversation, continued

E Conversations / turns 11-20

04

Continue with the word bank on page 37.

- 11 **Tessa:** Would this be an amendment, a new submission, or an exemption? Collaborators keep asking me which route applies, and I do not want to guess.
- 12 **Omar:** That requires an ⁶ _____. The designated office will assess the proposal under the applicable rules. We cannot announce the route before that assessment has been made.
- 13 **Tessa:** I can say an amendment might be needed, but none is approved. The draft wrongly implies approval is routine.
- 14 **Omar:** Keep any ⁷ _____ conditional. A plausible route is not a completed decision. Likewise, no waiver or new consent has been granted, so neither should appear as an established fact.
- 15 **Tessa:** The collaborator says they only want to check the available variables. Could I send a description of the proposed fields instead of the interviews?
- 16 **Omar:** Yes, describe the proposed fields without sharing participant information. The ⁸ _____ on the interviews remains. Do not promise that the files themselves will be released before the designated review.
- 17 **Tessa:** I will gather the approval, consent materials, key-access details, and proposed analysis description. That gives the office enough context to assess the new use.
- 18 **Omar:** Include the proposed ⁹ _____ arrangements if collaboration is planned. Such terms may be relevant, but they do not replace the required ethics determination or establish participant consent by themselves.
- 19 **Tessa:** My update will say the proposal is being prepared, no new analysis has begun, and collaborator access remains subject to the designated review.
- 20 **Omar:** That accurately describes the ¹⁰ _____. Keep preparation separate from conducting the new work. Once the authorized decision is recorded, follow its conditions rather than treating review as unlimited permission.

ANSWERS / EXPLANATIONS: PAGE 92

1. review status | 2. coded data | 3. linking key | 4. consent scope | 5. purpose change | 6. authorized determination | 7. protocol amendment | 8. access restriction | 9. data-use agreement | 10. secondary use

A pending route is not a permission

F Say it

A researcher proposes a new use of coded survey data. The team can access the linking key. The institution requires a designated-office determination before new analysis. The request has been sent but no decision issued.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

key / anonymous / determination / pending

Researcher: "The files are coded, not automatically ¹ _____."

Coordinator: "The proposed use still awaits an authorized ² _____."

Researcher: "Our team can access the linking ³ _____."

Coordinator: "We can accurately report that review is ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 92

1. anonymous | 2. determination | 3. key | 4. pending

Rehearse with a fixed script

1. Read turns 3-8, emphasizing that the research team can access the linking key.
2. Repeat turns 15-18, separating a general field description from sharing participant information.
3. Switch roles for the transfer and read the four completed lines without adding an approval claim.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

Credit follows contributions, not job titles

A Read the situation

05

Discuss authorship eligibility, outstanding responsibilities, and author order without granting or denying credit by status.

Research fellow Harper and project lead Marta discuss a manuscript whose target journal uses ICMJE authorship criteria: substantive research contribution, intellectual drafting or review, final-version approval, and accountability. All four are required. Harper designed and performed the main analysis but has not yet been invited to review the manuscript or approve the final version. Professor Nolan obtained funding only. The team has not agreed author order or documented contributions. Harper must have a genuine opportunity to complete the remaining responsibilities; funding alone does not establish Nolan's authorship. No final author list or order is settled.

PEOPLE / Harper: Research fellow | Marta: Project lead

DISCUSS CONTRIBUTION AND RESPONSIBILITY TOGETHER

Academic hierarchy can make credit conversations difficult. Refer to the agreed publication criteria and a factual contribution record. Do not frame authorship as a personal favor, and do not exclude a substantive contributor by withholding the chance to fulfill remaining responsibilities.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What substantive contribution has Harper made?

- A Already approved an unseen final manuscript
- C Designed and performed the main analysis

- B Determined author order alone
- D Obtained funding only

2. Does funding alone establish Nolan's authorship under the stated criteria?

- A Yes, if no contribution record exists
- C Yes, regardless of other work

- B No
- D Yes, if the title is professor

3. What should happen before the author list is finalized?

- A Guarantee Harper first position immediately.
- C Give Harper a genuine opportunity to review, approve, and accept the relevant responsibilities.

- B List everyone who attended a meeting.
- D Exclude Harper because no invitation was sent.

Answers and explanations: [page 93](#).

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

authorship

Credited responsibility for a published scholarly work.

IN USE establish authorship eligibility

contributorship

A description of what each person did.

IN USE document contributorship

byline

The author names listed on a publication.

IN USE finalize the byline

author order

The sequence in which authors are listed.

IN USE discuss author order

substantive contribution

Significant intellectual or research work on the project.

IN USE recognize a substantive contribution

conceptualization

Development of the study's ideas or research goals.

IN USE describe conceptualization

methodology

Development or design of research methods.

IN USE record methodology contributions

formal analysis

Application of analytical methods to study material or data.

IN USE credit formal analysis

data curation

Organizing and maintaining data for use.

IN USE describe data curation

investigation

Conducting the research or collecting relevant evidence.

IN USE record investigation work

funding acquisition

Obtaining financial support for a project.

IN USE acknowledge funding acquisition

supervision

Oversight or guidance of research work.

IN USE distinguish supervision from authorship

Precision check

Authorship eligibility and author order are separate decisions. A substantive analysis contribution matters, but the supplied journal requires all four responsibilities. Give Harper the opportunity to complete them; do not invent a final approval or a guaranteed first position.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

critical review

Intellectual examination of a manuscript's content.

IN USE contribute critical review

final approval

Agreement to the version intended for publication.

IN USE obtain final approval

accountability

Responsibility for addressing the work's accuracy and integrity.

IN USE accept accountability

corresponding author

The author managing the journal communication role.

IN USE identify the corresponding author

acknowledgment

Recognition of contributions outside the author byline.

IN USE agree an acknowledgment

contribution statement

A record describing each contributor's work.

IN USE prepare a contribution statement

authorship criteria

The applicable conditions for being listed as an author.

IN USE apply authorship criteria

honorary authorship

Listing someone without the required authorship basis.

IN USE avoid honorary authorship

omitted contributor

A contributor whose relevant work is not properly recognized.

IN USE address an omitted contributor

collaboration agreement

Recorded expectations for joint work.

IN USE revisit the collaboration agreement

dispute resolution

A process for addressing unresolved disagreement.

IN USE use the dispute-resolution process

manuscript version

An identified revision of a paper.

IN USE circulate the final manuscript version

Precision check

A contribution statement records work but does not automatically settle authorship. Under this case's adopted criteria, funding alone is insufficient. Apply the actual journal and institutional rules, since conventions differ by field.

Phrases you can actually use

C Notice the language / 1 of 2

OPEN FACTUALLY

"Let us document what each collaborator contributed."

USE THE AGREED STANDARD

"The target journal requires all four authorship criteria."

PROVIDE A REAL OPPORTUNITY

"Harper needs the manuscript and time to review it."

SEPARATE FUNDING

"Funding acquisition alone does not establish authorship here."

NAME THE ANALYTICAL WORK

"Harper designed and performed the main analysis."

IDENTIFY WHAT REMAINS

"The manuscript review and final approval have not happened yet."

AVOID EXCLUSION BY OMISSION

"Not receiving an invitation should not become a reason to deny credit."

KEEP TITLES OUT OF THE TEST

"Job title is not the criterion."

Contributed

Specify the actual work rather than relying on a title.

Eligible

Must be assessed against the applicable criteria.

First author

An ordering decision, not automatic from one criterion.

Phrases you can actually use

C Notice the language / 2 of 2

DISTINGUISH ORDERING

"Eligibility does not settle author order."

MAKE CREDIT VISIBLE

"Record contributions accurately in the statement."

IDENTIFY THE VERSION

"Approval must refer to the final manuscript version."

HANDLE DISAGREEMENT

"Use the institutional process if the group cannot resolve the dispute."

AVOID A PREMATURE PROMISE

"No final position has been agreed."

KEEP RESPONSIBILITY EXPLICIT

"Authorship includes accountability, not only recognition."

COORDINATE COMMUNICATION

"The corresponding-author role should be agreed."

CLOSE BEFORE SUBMISSION

"Settle contributions, responsibilities, and the author list before submitting."

Corresponding

A communication role, not necessarily the first-listed author.

Approved

Requires a real decision about an identified version.

Acknowledged

Recognition distinct from inclusion in the byline.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which statement about Harper is supported?

- A** Harper has a substantive analysis contribution and outstanding authorship responsibilities.
- B** Harper has already approved the final version.
- C** Harper did no relevant research work.
- D** Harper must automatically be first author.

2. Which action would unfairly block completion of the criteria?

- A** Circulating the manuscript for substantive review
- B** Recording the analysis contribution
- C** Discussing accountability explicitly
- D** Withholding the manuscript and then citing lack of review as a reason for exclusion

3. What does Nolan's funding-only role establish?

- A** Proof that Harper's analysis is irrelevant
- B** A contribution to record, but not sufficient authorship under the adopted criteria
- C** Automatic first authorship
- D** Final approval by every collaborator

4. What remains unresolved after identifying a substantive contribution?

- A** Whether the paper has a funding source
- B** Whether all titles are identical
- C** Whether the analysis ever occurred
- D** Completion of the remaining responsibilities and agreement on the author list and order

Answers and explanations: [page 93](#).

Credit follows contributions, not job titles

E Conversations / turns 1-10

05

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

author order / contribution statement / final approval / contributorship / dispute resolution / Accountability / critical review / funding acquisition / byline / corresponding author

- 01 **Harper:** My name was missing from yesterday's draft author list. I designed and performed the main analysis, but nobody discussed credit with me.
- 02 **Marta:** We should start with a ¹ _____. Your analytical work must be recorded accurately. The list is not final, and an informal draft should not substitute for a proper discussion with the collaborators.
- 03 **Harper:** Nobody sent me the manuscript. I need to check whether the methods and interpretation describe my work correctly.
- 04 **Marta:** You need a genuine opportunity for ² _____. I will circulate the manuscript and arrange time for your response. We should not withhold that opportunity and then cite its absence against you.
- 05 **Harper:** The journal uses ICMJE criteria, so I understand that doing the analysis is not the only responsibility. What still needs to happen before submission?
- 06 **Marta:** Beyond the substantive work and intellectual review, we need ³ _____ of the publication version and agreement on accountability. Those are real responsibilities, not boxes we can mark on someone's behalf.
- 07 **Harper:** I will review the full paper and discuss the analysis. Please do not claim my approval before that happens.
- 08 **Marta:** Agreed. ⁴ _____ must be understood and accepted. We will record the status honestly and identify outstanding steps, rather than inventing agreement to meet the submission timetable.
- 09 **Harper:** Professor Nolan is listed already, but his contribution so far was funding, without the other responsibilities.
- 10 **Marta:** We should record ⁵ _____ accurately too. Under the adopted criteria, that alone does not establish authorship. His title should neither replace the criteria nor prevent a respectful discussion of his actual contribution.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 46.

- 11 **Harper:** I also need us to discuss order. I had expected first authorship for the analysis work, but I realize we never agreed that as a group.
- 12 **Marta:** We must discuss ⁶ _____ separately. The criteria establish eligibility, not a universal ordering formula. We need a transparent conversation about the project and the journal's conventions, without promising a position prematurely.
- 13 **Harper:** Please invite the data collection and curation leads too. I do not want us settling my position without hearing what everyone else contributed.
- 14 **Marta:** Yes. Accurate ⁷ _____ requires the whole project's contribution record. We should not decide credit solely through the loudest objection, personal seniority, or who happened to see the draft first.
- 15 **Harper:** Who will communicate with the journal? People sometimes describe that role as though it always means first author.
- 16 **Marta:** The ⁸ _____ role should be agreed explicitly. It concerns communication and associated submission responsibilities. It does not automatically determine first position, and it cannot resolve an unsettled authorship dispute by itself.
- 17 **Harper:** If the group still cannot agree after reviewing contributions and responsibilities, we need a fair route forward rather than submitting an unresolved list.
- 18 **Marta:** Use the institutional ⁹ _____ process if necessary. We should document the issue and seek the appropriate assistance. A deadline does not justify bypassing a substantive disagreement about credit.
- 19 **Harper:** For now, please circulate the manuscript, record the analysis contribution, and arrange a group discussion. I will review the content before giving any final approval.
- 20 **Marta:** Then we can finalize the ¹⁰ _____ on an accurate basis before submission. No final list or order is settled today; the next step is to complete the review and resolve the outstanding decisions.

ANSWERS / EXPLANATIONS: PAGE 94

1. contribution statement | 2. critical review | 3. final approval | 4. Accountability | 5. funding acquisition | 6. author order | 7. contributorship | 8. corresponding author | 9. dispute resolution | 10. byline

An invitation is not an approval

F Say it

A journal uses the same four criteria. A collaborator made a substantive methods contribution and has just received the manuscript for review. No final approval or accountability agreement is yet recorded, and author order is undecided.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

critical review / outstanding / author order / contribution

Lead: "The methods work is a substantive ¹ _____."

Collaborator: "Receiving the manuscript gives me an opportunity for ² _____."

Lead: "Final approval remains ³ _____."

Collaborator: "Eligibility alone does not settle ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 94

1. contribution | 2. critical review | 3. outstanding | 4. author order

Rehearse with a fixed script

1. Read turns 1-8, keeping the contribution request direct and the remaining responsibilities explicit.
2. Repeat turns 9-16, distinguishing funding acquisition, author order, and corresponding-author duties.
3. Switch roles for the transfer; emphasize outstanding rather than implying final approval.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

Revise the claim, not the observed result

A Read the situation

Respond to a valid reviewer criticism with a precise revision and traceable manuscript locations.

06

Authors Mei and Luca respond to a reviewer of a fictional study involving 60 volunteer students at one campus. The study measured self-reported confidence after one workshop, with a mean of 7.1 on a ten-point scale. There was no baseline measurement, comparison group, or direct learning test. The manuscript nevertheless says the workshop improves learning for all adults. The authors agree to remove that claim from the abstract and conclusion, report the observed confidence in the studied sample, and explain the design limits. No new data or reanalysis has been performed.

PEOPLE / Mei: Corresponding author | Luca: Coauthor

AGREEMENT CAN BE SPECIFIC AND PROFESSIONAL

A reviewer response need not defend every original sentence. Acknowledge the valid point, say exactly what changed, and identify where it changed. Do not claim a limitation has been solved merely because the revised text now describes it honestly.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What did the study directly measure?

- A Long-term attainment in all adults
- B A randomized treatment effect
- C Self-reported confidence after one workshop
- D Improvement from a baseline learning test

2. Which original claim goes beyond the evidence?

- A The mean confidence rating was 7.1 out of ten.
- B No comparison group was included.
- C The workshop improves learning for all adults.
- D Sixty volunteer students participated at one campus.

3. What has been done for this revision?

- A A randomized trial has been completed.
- B Every adult population has been sampled.
- C The reported mean has been replaced by a more impressive number.
- D Wording is being corrected; no new data or reanalysis has occurred.

Answers and explanations: page 95.

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

peer review

Expert evaluation of scholarly work before or after publication.

IN USE respond to peer review

reviewer comment

A point raised by a manuscript evaluator.

IN USE address a reviewer comment

response letter

The document explaining replies and revisions.

IN USE prepare a response letter

point-by-point response

A reply organized against each review issue.

IN USE provide a point-by-point response

revision

A change to a manuscript or associated material.

IN USE document the revision

tracked changes

Visible marking of edits between document versions.

IN USE supply tracked changes

revised manuscript

The updated version submitted for consideration.

IN USE check the revised manuscript

abstract

A concise summary of a scholarly work.

IN USE revise the abstract

conclusion

The manuscript's final interpretation or takeaway.

IN USE narrow the conclusion

claim

An assertion about what the evidence establishes.

IN USE qualify a claim

overgeneralization

Extending a conclusion beyond its support.

IN USE remove an overgeneralization

sampling frame

The operational list or source of units available for selection, which may not fully cover the target population.

IN USE describe the sampling frame

Precision check

A post-workshop confidence mean describes a measured state, not improvement from baseline. Confidence and demonstrated learning are also different constructs. Changing the claim can correct reporting without altering the observed 7.1 mean.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

volunteer sample

Participants who chose to take part.

IN USE acknowledge the volunteer sample

single-site study

Research conducted at one location.

IN USE state the single-site limitation

self-report

Information supplied by participants about themselves.

IN USE distinguish self-report from direct measurement

post-only measure

An outcome recorded after an event without a baseline measure.

IN USE interpret a post-only measure

baseline measure

An observation collected before the relevant intervention or period.

IN USE identify the missing baseline measure

comparison condition

A reference condition for evaluating a contrast.

IN USE specify a comparison condition

construct

The concept a measure is intended to represent.

IN USE identify the measured construct

learning assessment

A task directly assessing relevant learning.

IN USE distinguish a learning assessment

limitation

A feature restricting interpretation or application.

IN USE state the study limitation

reanalysis

A new analysis of existing data.

IN USE distinguish revision from reanalysis

page-and-line reference

A precise locator for manuscript text.

IN USE provide page-and-line references

editorial decision

A journal editor's determination about the manuscript.

IN USE await the editorial decision

06

Precision check

The volunteer, single-campus sample limits broad application. An honest limitation statement does not repair the design or guarantee acceptance. Cite locations in the actual revised version, not stale page and line numbers from an earlier draft.

Phrases you can actually use

C Notice the language / 1 of 2

ACKNOWLEDGE THE POINT

"We agree that the original claim exceeded the design."

STATE THE POPULATION

"The sample comprised 60 volunteers at one campus."

REMOVE AN UNSUPPORTED CHANGE

"Without a baseline, we cannot claim improvement."

BOUND GENERALIZATION

"The result should not be stated for all adults."

IDENTIFY THE MEASURED CONSTRUCT

"We measured self-reported confidence, not demonstrated learning."

KEEP THE OBSERVATION

"The observed mean remains 7.1 out of ten."

REMOVE A CAUSAL CLAIM

"There was no comparison group to support that inference."

DESCRIBE THE REVISION

"We have narrowed the abstract and conclusion."

06

Improves

A change or effect claim requiring appropriate evidence.

Confidence

Not interchangeable with measured learning.

Volunteers

A sample feature relevant to interpretation.

Phrases you can actually use

C Notice the language / 2 of 2

MAKE THE LOCATION TRACEABLE

"The response identifies the revised page and lines."

SEPARATE HONESTY FROM REPAIR

"The limitation is now explicit, not eliminated."

CHECK CONSISTENCY

"The same claim boundary appears throughout the manuscript."

RESPECT THE DECISION

"The editor will assess the revised submission."

AVOID INVENTING WORK

"No new data collection or reanalysis was performed."

ANSWER THE COMMENT

"The revised wording reports the observation in the studied sample."

USE A CLEAN RESPONSE

"We state the change and its reason directly."

CLOSE THE ISSUE

"The response and manuscript now describe the same evidence."

06

Revised

Does not mean new data were collected.

Addressed

Can mean clarified, not that a design limitation disappeared.

Accepted

An editorial status, not guaranteed by agreement with a reviewer.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which revised claim is supported?

- A** Sixty volunteer students at one campus reported mean post-workshop confidence of 7.1 out of ten.
- B** The workshop raised confidence by 7.1 points.
- C** All adults learned more because of the workshop.
- D** Random assignment proved the effect.

2. Which response accurately describes the work?

- A** The abstract and conclusion are narrowed; no reanalysis was performed.
- B** New trial results solve the issue.
- C** The original mean is now invalid solely because the wording was too broad.
- D** The reviewer has guaranteed acceptance.

3. What does acknowledging the limitation accomplish?

- A** Makes a volunteer sample nationally representative
- B** Makes the interpretation more accurate without removing the design constraint
- C** Creates an unmeasured baseline
- D** Converts confidence to learning

4. Which locator should the response use?

- A** Only the phrase somewhere in the paper
- B** A fabricated appendix
- C** Page and line numbers verified against the revised manuscript
- D** Numbers copied without checking from an older version

Answers and explanations: [page 95](#).

Revise the claim, not the observed result

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

06

WORD + PHRASE BANK

revision / page-and-line reference / revised manuscript / post-only measure / reanalysis / editorial decision / overgeneralization / limitation / point-by-point response / construct

- 01 **Mei:** The reviewer says our claim about improving learning for all adults exceeds the sample and design. I agree, but the response needs to explain the correction precisely.
- 02 **Luca:** Start with the measured ¹ _____. We asked sixty volunteer students at one campus about confidence after the workshop. We did not directly assess learning, and the distinction belongs in the revised wording.
- 03 **Mei:** The mean was seven point one on a ten-point scale. That observed number can remain, provided we do not call it a seven-point improvement.
- 04 **Luca:** Right. It is a ² _____. We have no baseline measurement, so we cannot calculate a before-and-after change. The original sentence implied a comparison we never measured.
- 05 **Mei:** We also had no comparison group. Narrowing the claim should therefore address both the measured outcome and the lack of a design supporting that causal language.
- 06 **Luca:** State the ³ _____ directly. The study reports post-workshop confidence in this sample. It does not isolate the workshop's effect, and revising the wording does not change that design boundary.
- 07 **Mei:** What about all adults? Even if we had measured learning, sixty volunteers at one campus would not establish the same result in every adult population.
- 08 **Luca:** Remove that ⁴ _____. The revised statement should name the observed sample. We can discuss broader questions as future research without presenting them as conclusions already established by these participants.
- 09 **Mei:** The abstract and conclusion both contain the overclaim. We need to change both, not just the quoted sentence.
- 10 **Luca:** Check the ⁵ _____ throughout. The title, summary, results description, and final interpretation should not quietly restore the same unsupported claim after we have corrected it elsewhere.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 55.

- 11 **Mei:** I will quote the replacement sentence in our reply, then give its location. Could you check that the conclusion now says the same thing?
- 12 **Luca:** Use a ⁶ _____. For this comment, identify the outcome correction, the population boundary, and the design limitation. The reviewer should be able to verify each without guessing what we meant.
- 13 **Mei:** I will add the page and line numbers after the revised file is finalized. The current numbers could shift as we edit the abstract.
- 14 **Luca:** A checked ⁷ _____ matters. Use the submitted revision's locations, not an earlier draft's numbering. We should also keep the response wording consistent with the actual replacement text.
- 15 **Mei:** Someone suggested claiming additional analyses. We have not done them, and an impressive-sounding statement would not solve the problem.
- 16 **Luca:** Do not claim ⁸ _____. No new data or analysis was added. This revision corrects the interpretation and reporting of the existing observation; it does not create evidence the study never collected.
- 17 **Mei:** I have written limitation addressed. That sounds as though we fixed the study design. Would limitation clarified describe what we actually did?
- 18 **Luca:** Say the ⁹ _____ makes the boundary explicit. That is accurate and useful. We have improved the manuscript's claims, not repaired the completed design or established an effect through wording alone.
- 19 **Mei:** I will keep the mean, remove the improvement and all-adults claims, and verify locations before resubmission.
- 20 **Luca:** Then await the ¹⁰ _____. A clear correction supports evaluation, but it does not guarantee acceptance. Our responsibility is to make the evidence, changes, and remaining limitations straightforward to inspect.

06

ANSWERS / EXPLANATIONS: PAGE 96

1. construct | 2. post-only measure | 3. limitation | 4. overgeneralization | 5. revised manuscript | 6. point-by-point response | 7. page-and-line reference | 8. reanalysis | 9. revision | 10. editorial decision

A corrected claim, unchanged observations

F Say it

A single-campus survey records post-session satisfaction from 45 volunteers, with no baseline or comparison group. Authors remove a claim of improved performance. They change wording only and collect no new data.

06

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

satisfaction / eliminated / baseline / revision

Author: "The measured outcome was ¹ _____."

Reviewer: "The study cannot calculate improvement without a ² _____."

Author: "The work completed here is a wording ³ _____."

Reviewer: "The design limitation is explicit, not ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 96

1. satisfaction | 2. baseline | 3. revision | 4. eliminated

Rehearse with a fixed script

1. Read turns 1-8, distinguishing a post-only confidence score from improvement in learning.
2. Repeat turns 11-18, making the revision location and lack of reanalysis explicit.
3. Switch roles for the transfer and read explicit, not eliminated with a clear contrast.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

The missing-value code is not a temperature

A Read the situation

Make a dataset interpretable through verified units, missing-value rules, dates, and traceable processing.

Research data steward Elena helps analyst Ravi prepare a small environmental dataset. The column `t` contains 20, 22, -999, 24, and 26. The collector confirms that `t` is air temperature in degrees Celsius and -999 means a failed reading, not a temperature or zero. The date label 04/05/26 is confirmed as May 4, 2026; readings were taken at 09:00 UTC. The raw file must remain unchanged. An analysis copy will mark the failed reading as missing and calculate the mean of the four valid values. The release needs a data dictionary, processing record, and identified software version.

PEOPLE / Elena: Research data steward | Ravi: Analyst

07

MAKE THE FILE USABLE BY SOMEONE OUTSIDE THE TEAM

A familiar variable name can hide assumptions that a new collaborator cannot recover. Confirm definitions with the collector instead of guessing from plausible values. Document the transformation so a later reader can distinguish an observed value from a missing code and reproduce the analysis.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What does -999 represent in this dataset?

- | | |
|--|--|
| A A Fahrenheit value requiring conversion | B A failed reading coded as missing |
| C A valid Celsius temperature | D Exactly zero degrees |

2. What is the mean of the four valid temperatures?

- | | |
|---------------------------------|-------------------------------|
| A 92 degrees Celsius | B 23 degrees Celsius |
| C -181.4 degrees Celsius | D 18.4 degrees Celsius |

3. How should 04/05/26 be represented unambiguously after confirmation?

- | | |
|---------------------|---------------------|
| A 2026-05-26 | B 2004-05-26 |
| C 2026-05-04 | D 2026-04-05 |

Answers and explanations: [page 97](#).

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

data dictionary

Definitions and coding rules for dataset variables.

IN USE publish a data dictionary

metadata

Information describing data, collection, or processing.

IN USE supply essential metadata

variable name

The label assigned to a data field.

IN USE clarify variable names

unit of measurement

The scale used to express a quantity.

IN USE specify units of measurement

missing-value code

A designated value representing absent information.

IN USE document missing-value codes

sentinel value

A special value used to mark a status rather than an observation.

IN USE identify a sentinel value

failed reading

A measurement attempt without a valid recorded result.

IN USE mark a failed reading

valid observation

A recorded value meeting the applicable inclusion rules.

IN USE count valid observations

raw file

The original dataset before later cleaning or transformation.

IN USE preserve the raw file

analysis copy

A derived dataset used for analysis.

IN USE create a documented analysis copy

transformation log

A record of changes from source to derived data.

IN USE maintain a transformation log

data cleaning

Applying documented rules to correct or prepare data.

IN USE document data cleaning

Precision check

The valid values are 20, 22, 24, and 26: their mean is 23 degrees Celsius. Treating -999 as a temperature would produce -181.4. Replacing it with zero would produce 18.4. Neither represents the stated valid-observation calculation.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

imputation

Replacing missing values using a specified estimation method.

IN USE distinguish cleaning from imputation

complete-case calculation

A calculation using records with required observed values.

IN USE describe a complete-case calculation

date format

The convention used to represent a calendar date.

IN USE standardize the date format

time zone

The reference used to interpret recorded times.

IN USE record the time zone

collection timestamp

The recorded date and time of observation.

IN USE preserve collection timestamps

version identifier

A label identifying a particular dataset or software revision.

IN USE record the version identifier

software environment

The tools and versions used to run an analysis.

IN USE document the software environment

dependency

A required library or component for a process.

IN USE list analysis dependencies

readme

An introductory file explaining a dataset or project.

IN USE include a readme

repository

A managed location for preserving and accessing materials.

IN USE select a suitable repository

computational reproducibility

Reobtaining results with the specified data and computation.

IN USE check computational reproducibility

data validation

Checks that data meet specified rules or expectations.

IN USE perform data validation

Precision check

The analysis uses four observed values; it does not estimate the failed reading. Preserve the raw sentinel and document its conversion to missing in the analysis copy. The collector, not a guess about regional date conventions, resolves the date.

Phrases you can actually use

C Notice the language / 1 of 2

ASK FOR MEANING

"What does the variable t represent?"

IDENTIFY THE SPECIAL CODE

"Minus 999 marks a failed reading."

KEEP THE ORIGINAL

"The raw file remains unchanged."

STATE THE CALCULATION BASE

"The mean uses four valid observations."

CONFIRM THE UNIT

"The collector confirms degrees Celsius."

REJECT A FALSE ZERO

"Missing is not the same as zero."

DESCRIBE THE DERIVED COPY

"The analysis copy marks that code as missing."

GIVE THE RESULT WITH UNITS

"The valid-observation mean is 23 degrees Celsius."

07

Plausible

Not a substitute for a verified variable definition.

Missing

An absent valid observation, not zero.

Excluded from this mean

Does not mean erased from the raw record.

Phrases you can actually use

C Notice the language / 2 of 2

AVOID INVENTED DATA

"We have not imputed the failed reading."

USE AN UNAMBIGUOUS FORMAT

"Record the date as 2026-05-04."

DOCUMENT THE FIELDS

"The data dictionary must include units and missing codes."

IDENTIFY THE ENVIRONMENT

"State the software version used for the analysis."

RESOLVE THE DATE

"The collector confirms May 4, 2026."

KEEP TIME MEANINGFUL

"The collection time is 09:00 UTC."

RECORD THE TRANSFORMATION

"Log the conversion and the calculation rule."

CHECK THE RELEASE

"Another analyst should be able to reproduce the reported mean."

07

Imputed

An estimated replacement; none is supplied in this case.

May 4

Confirmed meaning of the otherwise ambiguous source label.

Reproducible

Computation can repeat; that alone does not prove scientific validity.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which data-handling description is correct?

- A** Overwrite the raw value without a record.
- B** Replace the failed reading with zero and call it observed.
- C** Treat -999 as a valid air temperature.
- D** Preserve the raw file and convert the sentinel to missing in a documented analysis copy.

2. What does using four valid observations imply?

- A** The missing temperature is proved to be zero.
- B** Five valid temperatures were measured.
- C** The raw file must be deleted.
- D** No value has been imputed for the failed reading.

3. Why is the collector's date confirmation necessary?

- A** The original numeric label could otherwise be interpreted in different date orders.
- B** All countries use the same numeric date order.
- C** Time zone alone decides the calendar format.
- D** The date should be chosen to improve the mean.

4. Which release package best supports reproducibility?

- A** A screenshot with an unexplained missing code
- B** Data, dictionary, processing steps, and the identified software environment
- C** Only a column named t
- D** Only the final mean without units

[Answers and explanations: page 97.](#)

The missing-value code is not a temperature

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

unit of measurement / transformation log / date format / time zone / computational reproducibility / raw file / imputation / software environment / valid observations / missing-value code

07

- 01 **Ravi:** The shared file has a column called t and five values. Four look like ordinary temperatures, but the fifth is minus nine hundred ninety-nine.
- 02 **Elena:** Do not guess the ¹ _____ from the values. The collector confirms air temperature in degrees Celsius. We need that definition in the documentation, because the column name alone does not establish it.
- 03 **Ravi:** The collector confirmed that minus nine hundred ninety-nine codes a failed reading, not a cold observation.
- 04 **Elena:** Document the ² _____ and mark it missing in the analysis copy. Do not use zero: that would turn a failed reading into an invented temperature.
- 05 **Ravi:** Including the code gives a mean of minus one hundred eighty-one point four, which is not the supported calculation.
- 06 **Elena:** Use the four ³ _____ for this mean: twenty, twenty-two, twenty-four, and twenty-six. They sum to ninety-two, so the mean is twenty-three degrees Celsius. State the four-observation base alongside it.
- 07 **Ravi:** Replacing the code with zero would give eighteen point four instead. That is also wrong for this requested mean, even though it looks less obviously unusual.
- 08 **Elena:** Correct. We are not performing ⁴ _____. We have no estimated replacement for the failed reading. The analysis should say exactly which observed values were used, without turning an absent measurement into a convenient number.
- 09 **Ravi:** I can make a separate cleaned file. Should the original keep minus nine hundred ninety-nine so the transformation can be checked later?
- 10 **Elena:** Preserve the ⁵ _____. Keep the original code and create a documented analysis copy. Future users need both the source record and the explanation of how the derived data were produced.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 64.

- 11 **Ravi:** There is also a date labeled zero-four slash zero-five slash twenty-six. The collector says it means May fourth, twenty twenty-six, not April fifth.
- 12 **Elena:** Use an unambiguous ⁶ _____ such as 2026-05-04 in the documented copy. Record how you resolved the source label. A plausible regional assumption would not be enough without that confirmation.
- 13 **Ravi:** The readings were taken at nine in the morning UTC. I will include that reference, because nine o'clock alone could mean different local times.
- 14 **Elena:** Yes, the ⁷ _____ belongs with the collection information. Keep the timestamp meaning explicit so another researcher does not align these readings with observations from a different hour.
- 15 **Ravi:** The data dictionary will define t, the Celsius unit, the failed-reading code, and the confirmed date and time conventions. What else belongs in the release?
- 16 **Elena:** Add a ⁸ _____ describing the missing-code conversion and mean calculation. It should connect the raw file to the analysis copy and reported result without requiring a colleague to reconstruct undocumented choices.
- 17 **Ravi:** I will add the software version and dependencies to the readme. Could you check that the instructions start from the raw file?
- 18 **Elena:** Document the ⁹ _____ with the data and instructions. Even a simple calculation benefits from a traceable version, especially when this small example becomes part of a larger processing workflow.
- 19 **Ravi:** I will ask another analyst to follow the release instructions and reproduce the twenty-three-degree mean from the preserved source and documented rules.
- 20 **Elena:** That checks ¹⁰ _____. A matching result is useful, but it does not prove scientific validity. Keep the variable definitions with the calculation so readers can check both.

07

ANSWERS / EXPLANATIONS: PAGE 98

1. unit of measurement | 2. missing-value code | 3. valid observations | 4. imputation | 5. raw file | 6. date format | 7. time zone | 8. transformation log | 9. software environment | 10. computational reproducibility

Do not turn missing into zero

F Say it

A confirmed Celsius column contains 10, 12, -999, and 14. The collector defines -999 as missing. The requested mean uses only observed temperatures, and the original file must be preserved.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

imputed / three / original file / 12

Analyst: "There are ¹ _____ valid observations."

Steward: "Their mean is ² _____ degrees Celsius."

Analyst: "The raw sentinel is retained in the ³ _____."

Steward: "No missing temperature has been ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 98

1. three | 2. 12 | 3. original file | 4. imputed

Rehearse with a fixed script

1. Read turns 1-8, saying minus nine hundred ninety-nine as a code and twenty-three degrees Celsius as the mean.
2. Repeat turns 11-14 with May fourth and UTC clearly audible.
3. Switch roles for the transfer; distinguish three observations from the mean of twelve.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

Answer the challenge without claiming universal proof

A Read the situation

Give a concise evidence-based conference answer, distinguish task performance from mechanism, and handle an unresolved question.

Presenter Noor answers a conference question about a memory-strategy study. All 24 volunteers completed one laboratory task in a fixed before-then-after order. Mean response time was 600 milliseconds before training and 540 afterward, a 60-millisecond or 10% reduction. There was no untrained comparison group, so practice and order effects are not isolated from training. No confidence interval or significance test is supplied for this exercise. An audience member asks whether the study proves a general memory theory. During the follow-up discussion, Noor must answer concisely, stating observations, limits, and the next test without inventing evidence.

PEOPLE / Audience member: Conference participant | Noor: Presenter

08

ANSWER THE STRONGEST REASONABLE VERSION OF THE QUESTION

A challenging question can help clarify the contribution. Give the direct answer first, then the main evidence and limitation. You can reject an overstatement without dismissing your own result or treating the questioner as hostile. Offer a relevant next study rather than an unsupported promise.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What change was observed in mean response time?

- | | |
|--|---|
| A A reduction of 540 milliseconds, or 90% | B A 60% increase |
| C No before-and-after difference | D A reduction of 60 milliseconds, or 10% |

2. Which alternative explanations are not isolated?

- | | |
|---|--|
| A Randomized group allocation | B Practice and order effects |
| C A confirmed change in all adult memory systems | D A published confidence interval |

3. What should Noor say about a confidence interval?

- | | |
|---------------------------------------|--|
| A It must be 95% to 100%. | B None is supplied here, so no interval should be invented. |
| C It is exactly zero to sixty. | D It proves the general theory. |

Answers and explanations: page 99.

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

conference presentation

A spoken account of research for a scholarly audience.

IN USE deliver a conference presentation

audience question

A query raised by someone attending a talk.

IN USE answer an audience question

central finding

The main observation or result being reported.

IN USE state the central finding

take-home message

The concise main point an audience should retain.

IN USE clarify the take-home message

response time

The elapsed time to produce a response in a task.

IN USE measure response time

millisecond

One thousandth of a second.

IN USE report milliseconds

within-participant comparison

A comparison of observations from the same participants.

IN USE describe a within-participant comparison

fixed order

The same sequence of conditions for all participants.

IN USE acknowledge a fixed order

practice effect

A performance change associated with repeated task exposure.

IN USE consider a practice effect

order effect

A change associated with the sequence of conditions.

IN USE control for order effects

comparison arm

A study group or condition serving as a reference.

IN USE include a comparison arm

counterbalancing

Varying condition order across participants under a design.

IN USE plan appropriate counterbalancing

Precision check

The mean fell from 600 to 540 milliseconds: 60/600 equals 10%. Lower response time is faster performance on this task. That observation alone does not identify the mechanism, isolate training from practice, or establish a universal memory theory.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

mechanism

The process proposed to explain an observed effect.

IN USE test a proposed mechanism

theoretical claim

An assertion about a broader explanatory account.

IN USE bound a theoretical claim

generalizability

The extent to which results apply beyond the study context.

IN USE assess generalizability

statistical uncertainty

Uncertainty associated with estimation from data.

IN USE communicate statistical uncertainty

confidence interval

An interval produced by a method with stated long-run coverage.

IN USE report a confidence interval

significance test

A statistical assessment under a specified null model.

IN USE interpret a significance test

independent replication

A new study testing a finding with independent evidence.

IN USE seek independent replication

boundary condition

A circumstance limiting when a finding or explanation applies.

IN USE identify boundary conditions

controlled follow-up

A later study designed to address relevant alternative explanations.

IN USE propose a controlled follow-up

clarifying question

A question used to resolve ambiguity in what is being asked.

IN USE ask a clarifying question

time allocation

The speaking time available for a response.

IN USE respect the time allocation

unresolved question

An issue not answered by the current evidence.

IN USE state the unresolved question

Precision check

A sample size and two means do not supply a confidence interval or significance test by themselves. Do not invent either. State the available evidence and propose a design that addresses the fixed order, absent comparison group, and narrow setting.

Phrases you can actually use

C Notice the language / 1 of 2

ANSWER DIRECTLY

"No, this study does not prove the general theory."

GIVE THE RELATIVE CHANGE

"That is a 10% reduction from the initial mean."

SPECIFY THE SETTING

"This was one laboratory task."

NAME THE ALTERNATIVE

"Practice and order effects remain possible explanations."

STATE THE OBSERVED CHANGE

"Mean response time fell from 600 to 540 milliseconds."

NAME THE SAMPLE

"All 24 volunteers completed the task."

DESCRIBE THE DESIGN

"Everyone completed the same before-then-after sequence."

AVOID A CAUSAL SHORTCUT

"We have not isolated training from those effects."

08

Proves

Often stronger than a single empirical study warrants.

Faster

A task-performance description, not a complete explanation.

Before and after

Does not by itself isolate the intervention.

Phrases you can actually use

C Notice the language / 2 of 2

SEPARATE MECHANISM

"Faster responses do not establish the proposed memory mechanism."

KEEP THE CONTRIBUTION

"The observation is worth testing under stronger controls."

BOUND GENERALIZATION

"Broader populations and tasks need separate evidence."

RESPECT THE TIME

"The short answer is that the finding is limited but informative."

HANDLE MISSING STATISTICS

"I do not have a verified interval or test result to report here."

PROPOSE THE NEXT TEST

"A controlled follow-up should address the alternative explanations."

CLARIFY THE QUESTION

"Are you asking about the observed change or the proposed mechanism?"

CLOSE WITHOUT OVERPROMISING

"The next study would test the explanation, not guarantee confirmation."

08

Could reflect

Marks an alternative explanation rather than a finding.

Not supplied

Do not invent a statistic to fill the conversational gap.

Next test

A proposal, not an already completed replication.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which opening best answers the audience's question?

- A "The question is hostile and cannot be answered."
- B "The effect is definitely statistically significant."
- C "No; the study reports a task-specific change, not proof of the general theory."
- D "Yes; every theory is proved by one lower mean."

2. Which causal claim is unsupported?

- A The study used 24 volunteers.
- B Training alone caused the entire reduction.
- C Mean response time was lower afterward.
- D All participants followed the same sequence.

3. Which next step best addresses the limitation?

- A Hiding the fixed sequence
- B Calling the same data an independent replication
- C A controlled follow-up designed to separate training from practice and order effects
- D Repeating the universal claim more confidently

4. Which statement about the mechanism is accurate?

- A Lower response time identifies every internal memory process.
- B The mechanism was directly measured because the talk was accepted.
- C A proposed theory and an observed mean are identical evidence.
- D The observed response-time change does not by itself establish the memory mechanism.

Answers and explanations: [page 99](#).

Answer the challenge without claiming universal proof

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

controlled follow-up / response time / within-participant comparison / Generalizability / practice effect / mechanism / take-home message / confidence interval / unresolved question / central finding

08

- 01 **Audience member:** You showed a faster response after training. Does that prove the general memory theory, or are you making a narrower claim about this task?
- 02 **Noor:** Our ¹ _____ is narrower. Mean response time fell from six hundred to five hundred forty milliseconds. This does not prove the general theory.
- 03 **Audience member:** What percentage is that reduction? We need the initial response time to interpret its relative size.
- 04 **Noor:** The ² _____ reduction is ten percent of the initial mean. All twenty-four volunteers completed the task. That calculation describes the observed change, without explaining its cause or importance beyond this setting.
- 05 **Audience member:** Did different groups perform the before and after conditions, or were these repeated observations from the same people?
- 06 **Noor:** It was a ³ _____. Everyone completed the before condition and then the after condition in the same order. That controls neither every time-related influence nor the effect of repeating the task.
- 07 **Audience member:** People may be faster through practice. How did you separate that possibility from training?
- 08 **Noor:** We did not isolate the ⁴ _____. There was no untrained comparison group. Repetition may contribute to the reduction, and the fixed order leaves related explanations open. I cannot attribute the entire change to training alone.
- 09 **Audience member:** Did you measure the memory process itself, or is it a proposed explanation for faster responses?
- 10 **Noor:** It remains a proposed ⁵ _____. Faster responses do not identify the underlying process. This observation motivates testing that explanation; it cannot replace such a test.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 73.

- 11 **Audience member:** What confidence interval surrounds the reduction? It would help us judge how much uncertainty there is around the estimated change.
- 12 **Noor:** I do not have a verified ⁶ _____ to report here, and I should not guess. The sample size and two means alone do not provide enough information for a defensible interval calculation.
- 13 **Audience member:** Fair enough. Would a controlled follow-up be aimed at repeating the same observation, or at separating the competing explanations?
- 14 **Noor:** A ⁷ _____ should address those alternatives explicitly. The design needs an appropriate comparison and attention to sequence, with its analysis planned before results are examined. That is proposed work, not a completed validation.
- 15 **Audience member:** Even if that study supports a training effect, would you still need evidence beyond this task and these volunteers before making the broad theory claim?
- 16 **Noor:** Yes. ⁸ _____ remains a separate issue. Other populations, tasks, and conditions could behave differently. A better-controlled local result would not automatically establish every application of the theory.
- 17 **Audience member:** The chair is asking us to wrap up. Could you give us the main result in one sentence, with that qualification?
- 18 **Noor:** The ⁹ _____ is this: response time fell ten percent in twenty-four volunteers, but we have not separated the training effect from practice and order effects.
- 19 **Audience member:** Thank you. That distinguishes the observed change from a general explanation. The planned test sounds like the important next step.
- 20 **Noor:** Exactly. The ¹⁰ _____ is what produced the change and how broadly it applies. We will keep those questions explicit rather than promise that the next study must confirm our preferred explanation.

08

ANSWERS / EXPLANATIONS: PAGE 100

1. central finding | 2. response time | 3. within-participant comparison | 4. practice effect | 5. mechanism | 6. confidence interval | 7. controlled follow-up | 8. Generalizability | 9. take-home message | 10. unresolved question

Give the observation and its boundary

F Say it

A fixed-order task study records mean times of 800 milliseconds before training and 720 afterward in 16 volunteers. There is no untrained comparison group, and no interval or significance result is supplied.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

80 / invented / 10 / practice and order effects

Presenter: "The mean reduction is ¹ _____ milliseconds."

Questioner: "Relative to the initial mean, that is ² _____ percent."

Presenter: "Training has not been isolated from ³ _____."

Questioner: "A significance result must not be ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 100

1. 80 | 2. 10 | 3. practice and order effects | 4. invented

Rehearse with a fixed script

1. Read turns 1-8, giving the direct answer before the sample and design qualification.
2. Repeat turns 11-14 without inventing a confidence interval; keep the follow-up study in the future tense.
3. Switch roles for the transfer and read eighty milliseconds and ten percent as different descriptions of the same change.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

The smallest p-value was not the primary outcome

E More workplace conversations / 1-10

Distinguish a prespecified analysis from a result selected after inspection.

Before collecting data, a fictional team registered one primary outcome and a 0.05 significance threshold. Its planned test gives $p = 0.18$. Afterward, the team explores twenty other outcomes; the smallest unadjusted p-value is 0.03. No multiple-testing adjustment or independent confirmation has been performed.

PEOPLE / Alina: Postdoctoral researcher | Dev: Statistical collaborator

Use each bank entry once, without changing its form. Answers follow turn 20.

09

WORD + PHRASE BANK

multiple testing / exploratory / deviation statement / p-value / primary outcome / independent confirmation

- 01 **Alina:** The main test was not significant, but I found an outcome with p point zero three. Could we lead with that one instead?
- 02 **Dev:** We can report it, but not replace the ¹ _____ in the account of what we planned. The registered analysis still belongs in the results.
- 03 **Alina:** Its p-value is point one eight against our point zero five threshold. I will show that result rather than remove the table.
- 04 **Dev:** Good. Say that the planned test did not meet that threshold, not that we proved there is no effect.
- 05 **Alina:** For the other outcome, I was going to write that our hypothesis was confirmed. We did discuss it informally before the project.
- 06 **Dev:** This particular test came after inspection. Label it ² _____. An earlier informal discussion is not evidence that this analysis was the registered test.
- 07 **Alina:** Should we omit the exploratory work altogether? The pattern might help us design the next study.
- 08 **Dev:** Keep useful exploration. Be clear about the sequence of decisions and what the analysis can support. Transparency does not mean pretending discovery has no value.
- 09 **Alina:** The draft lists only the outcome with point zero three. The other nineteen are in my notebook but not in the manuscript.
- 10 **Dev:** That hides the ³ _____ issue. Describe the family of analyses, the selection, and that the reported p-value is unadjusted.

The conversation, continued

E More workplace conversations / 11-20

Continue with the word bank on page 76.

- 11 **Alina:** Could I call point zero three a ninety-seven percent probability that the hypothesis is correct? That wording would be easier for the audience.
- 12 **Dev:** No. A ⁴ _____ is not the probability that our hypothesis is true. We need the estimate, uncertainty, and modeling assumptions, not that conversion.
- 13 **Alina:** We have the estimate in the output. I will check its units and the interval calculation with you before using either in the summary.
- 14 **Dev:** Yes. And keep the exploratory status visible even if a later analysis uses a different adjustment. A new calculation does not change when we selected the outcome.
- 15 **Alina:** For the follow-up, could we specify this outcome and analysis before collecting new data?
- 16 **Dev:** That would support an ⁵ _____ attempt. Define the design and sample-size rationale first; do not promise the new result will agree.
- 17 **Alina:** I will also link the original registration. There was one exclusion change during analysis that the methods section does not currently explain.
- 18 **Dev:** Document the original rule, what changed, when, and why. Readers need to distinguish that deviation from the planned analysis rather than infer it from mismatched sample counts.
- 19 **Alina:** Then we retain the primary result, report the twenty exploratory tests transparently, and propose a separately planned follow-up.
- 20 **Dev:** Exactly. Add a ⁶ _____ for the exclusion change. We can present a useful finding without rewriting the study's history.

09

ANSWERS / EXPLANATIONS: PAGE 101

1. primary outcome | 2. exploratory | 3. multiple testing | 4. p-value | 5. independent confirmation | 6. deviation statement

Keep the planned and discovered results separate

F Say it in a new situation

A registered primary test has $p = 0.12$ against a 0.05 threshold. Twelve unplanned outcomes are explored; the smallest unadjusted p-value is 0.02. The same dataset is used throughout.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

exploratory / threshold / confirmation / unadjusted

09

Researcher: "The registered test did not meet the specified ¹ _____."

Collaborator: "The twelve later tests are ² _____."

Researcher: "The reported minimum p-value is ³ _____."

Collaborator: "Reusing these inspected data is not independent
⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 101

1. threshold | 2. exploratory | 3. unadjusted | 4. confirmation

Read, switch, repeat

1. Check the four answers and read the complete exchange aloud.
2. Switch roles. Read it again with the word bank covered.
3. Return to the extended conversation. Read turns 1-10, then switch roles for turns 11-20. Use the corrected expressions.

SOLO PRACTICE

Read both roles. Pause briefly at each change of speaker. Repeat any line that needed an answer check.

The indirect-cost rate applies to the base, not every line

E More workplace conversations / 1-10

Read back a research budget while distinguishing cost totals, exclusions, and commitments.

A fictional foundation permits \$70,000 total costs. Agreed direct lines are personnel including benefits \$40,000, supplies \$8,000, equipment \$12,000, and travel \$4,000. Its specific terms allow 10% indirect costs on direct costs excluding equipment. These are supplied grant terms, not a general funding rule. Internal submission approval remains pending.

PEOPLE / Mina: Principal investigator | Reuben: Sponsored-projects officer

Use each bank entry once, without changing its form. Answers follow turn 20.

10

WORD + PHRASE BANK

total request / cost sharing / cost base / indirect costs / budget justification / personnel total

- 01 **Mina:** My budget totals sixty-four thousand in direct costs. I added ten percent to that and got seventy thousand four hundred overall.
- 02 **Reuben:** The addition is arithmetically right, but it uses the wrong ¹ _____. This foundation excludes equipment before applying the indirect-cost rate.
- 03 **Mina:** So the twelve-thousand equipment line is removed from the calculation, not removed from the amount we are requesting.
- 04 **Reuben:** Exactly. The equipment remains a direct project cost. Subtract twelve thousand from sixty-four thousand only to calculate the eligible base.
- 05 **Mina:** That leaves fifty-two thousand. Ten percent is five thousand two hundred. I had overstated the indirect amount by twelve hundred.
- 06 **Reuben:** Correct. Those are the ² _____ under these particular terms. Use the foundation's rule here, not a rate copied from another award.
- 07 **Mina:** Then the full request is sixty-four thousand plus five thousand two hundred, or sixty-nine thousand two hundred.
- 08 **Reuben:** Yes. That is eight hundred below the seventy-thousand total-cost ceiling. It is not eight hundred of money already awarded or available to spend.
- 09 **Mina:** Should I use the spare eight hundred as a contingency line? There is no separate contingency category in this call.
- 10 **Reuben:** Do not add an unsupported ³ _____. We should request supported project costs, not fill the ceiling merely because space remains.

The conversation, continued

E More workplace conversations / 11-20

Continue with the word bank on page 79.

- 11 **Mina:** For personnel, the forty thousand already includes benefits. I see another benefits percentage on the spreadsheet template. That would count them twice.
- 12 **Reuben:** Remove that duplicate addition. The ⁴ _____ is the agreed forty thousand. Make the inclusion explicit in the explanatory note.
- 13 **Mina:** The foundation template also asks for effort. My commitment is twenty percent of a twelve-month appointment for this project year.
- 14 **Reuben:** That is two point four person-months. Effort describes time committed; it is not automatically an additional budget line beyond the personnel costs already calculated.
- 15 **Mina:** I will show two point four person-months and explain the salary calculation separately. We are not proposing an unpaid contribution in this budget.
- 16 **Reuben:** Good. Do not label the difference below the ceiling ⁵ _____. An unused ceiling is not an institutional promise to fund part of the project.
- 17 **Mina:** The totals now reconcile. Can I press submit, or is the internal approval still needed even though we are below the maximum?
- 18 **Reuben:** It is still needed. I will send the corrected budget for the authorized review. This calculation does not commit the university or confirm an award.
- 19 **Mina:** I will keep the submission pending and attach the foundation's costing terms, the line-item budget, and the explanations.
- 20 **Reuben:** Then the ⁶ _____ is sixty-nine thousand two hundred, subject to internal approval. The direct costs, indirect base, rate, and overall ceiling should each remain visible.

10

ANSWERS / EXPLANATIONS: PAGE 102

1. cost base | 2. indirect costs | 3. budget justification | 4. personnel total | 5. cost sharing | 6. total request

Exclude from the base, retain in the request

F Say it in a new situation

A fictional grant has \$50,000 direct costs, including \$10,000 equipment. Its terms permit 15% indirect costs on direct costs excluding equipment. The total-cost ceiling is \$60,000. No award has been made.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

56,000 / 6,000 / 40,000 / 4,000

10

Investigator: "The indirect-cost base is ¹ _____ dollars."

Officer: "The indirect-cost amount is ² _____ dollars."

Investigator: "The combined request is ³ _____ dollars."

Officer: "The unused ceiling is ⁴ _____ dollars, not an award."

ANSWERS / EXPLANATIONS: PAGE 102

1. 40,000 | 2. 6,000 | 3. 56,000 | 4. 4,000

Read, switch, repeat

1. Check the four answers and read the complete exchange aloud.
2. Switch roles. Read it again with the word bank covered.
3. Return to the extended conversation. Read turns 1-10, then switch roles for turns 11-20. Use the corrected expressions.

SOLO PRACTICE

Read both roles. Pause briefly at each change of speaker. Repeat any line that needed an answer check.

Which manuscript can the repository release?

E More workplace conversations / 1-10

Identify an article version and explain a specific deposit condition without inventing broader rights.

For a fictional article published January 15, 2026, the checked agreement permits repository deposit of the accepted manuscript now, but public release only from October 15, 2026. It does not permit deposit of the publisher PDF. Today is October 1. No conflicting funder requirement applies in this case.

PEOPLE / Nadia: Researcher | Ellis: Repository librarian

Use each bank entry once, without changing its form. Answers follow turn 20.

WORD + PHRASE BANK

metadata / embargo / version of record / release conditions / accepted manuscript / third-party material

11

- 01 **Nadia:** I have the journal PDF and would like it on my university profile today. Is that the file I should upload to the repository?
- 02 **Ellis:** Not under this agreement. We need the ¹ _____, including the changes accepted by the journal, before the publisher's typesetting.
- 03 **Nadia:** There are three files in our folder: the original submission, the accepted Word file, and the formatted journal PDF. I can supply the middle one.
- 04 **Ellis:** Please do. Check that it is the actual accepted text, not a later personal edit or a draft that predates the reviewers' changes.
- 05 **Nadia:** The formatted file has the final page numbers and logo. Would using it make the record more accurate?
- 06 **Ellis:** We can link to the ² _____ through its identifier. That does not give us permission to deposit the publisher PDF when this agreement excludes it.
- 07 **Nadia:** The article was published January fifteenth. The agreement lists October fifteenth as the public-release date. Today is October first, so I thought we were close enough.
- 08 **Ellis:** The date has not arrived. The checked terms allow deposit now with access closed, then public release on October fifteenth. We should not round the date down.
- 09 **Nadia:** So an item can be in the repository without its full text already being available to every visitor?
- 10 **Ellis:** Yes. The ³ _____ controls the full-text release. We will configure the record according to these specific terms rather than assume every deposit is immediately open.

The conversation, continued

E More workplace conversations / 11-20

Continue with the word bank on page 82.

- 11 **Nadia:** I have also attached a chart reproduced from another publisher. The article agreement does not say whether that chart can be included in repository copies.
- 12 **Ellis:** Flag the ⁴ _____ for a rights check before release. We should not infer its status from permission covering your own manuscript text.
- 13 **Nadia:** I can send the chart permission with the agreement. If that permission is limited, you will tell me what version can lawfully be made available?
- 14 **Ellis:** Yes, we will establish the permitted route with the relevant support team. Do not silently alter the accepted paper and label the result as the unmodified accepted version.
- 15 **Nadia:** For the repository description, I will add the article title, author list, journal reference, and persistent identifier.
- 16 **Ellis:** Those ⁵ _____ help readers identify the work. We will also label the deposited file's version clearly so they know what they are opening.
- 17 **Nadia:** Can I add a Creative Commons license to make it easier for people to reuse? I do not see one granted in the documents we have checked.
- 18 **Ellis:** Not on that basis alone. We need authority for the license applied to the deposited material. Availability and permission to redistribute are not the same thing.
- 19 **Nadia:** I will supply the accepted file and chart permission today. The public full text stays closed until the date and the rights check are satisfied.
- 20 **Ellis:** Correct. I will record the ⁶ _____. We can prepare the record now without promising public access before those conditions are met.

11

ANSWERS / EXPLANATIONS: PAGE 103

1. accepted manuscript | 2. version of record | 3. embargo | 4. third-party material | 5. metadata | 6. release conditions

Deposit permission has a version and a date

F Say it in a new situation

A fictional checked agreement allows the accepted manuscript to be deposited now and released publicly from December 1. Today is November 20. The publisher PDF is excluded, and no third-party permissions are unresolved.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

December 1 / embargo / manuscript / excluded

11

Researcher: "The permitted file is the accepted ¹ _____."

Librarian: "Public full-text access remains closed during the ² _____."

Researcher: "The earliest stated public-release date is ³ _____."

Librarian: "The publisher PDF remains ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 103

1. manuscript | 2. embargo | 3. December 1 | 4. excluded

Read, switch, repeat

1. Check the four answers and read the complete exchange aloud.
2. Switch roles. Read it again with the word bank covered.
3. Return to the extended conversation. Read turns 1-10, then switch roles for turns 11-20. Use the corrected expressions.

SOLO PRACTICE

Read both roles. Pause briefly at each change of speaker. Repeat any line that needed an answer check.

01 Briefing and language checks

Research Questions and Study Design

A Briefing checks / page 4

1. What is the completion rate among tutoring participants?

A. 80%

Thirty-two completers divided by forty participants equals eighty percent.

2. What is the difference between the two completion rates?

B. Twenty percentage points

Participant completion is eighty percent and nonparticipant completion is sixty percent, a twenty-point difference.

3. Why is a causal claim unsupported here?

B. Tutoring was self-selected and relevant group differences are unmeasured.

Self-selection and unmeasured differences prevent the observed contrast from isolating tutoring's causal effect.

D Language checks / page 9

1. Which headline fits the available evidence?

A. Tutoring participation was associated with higher completion in this course cohort.

The supported headline preserves an association; the alternatives add causation, misstate relative change, or confuse sampling with assignment.

2. Which question is still unresolved?

C. Whether tutoring caused the completion difference

The current observational records do not isolate the causal contribution of tutoring.

3. Which statement about possible confounding is accurate?

D. Prior preparation or motivation may differ, but these records do not establish those differences.

The brief identifies plausible unmeasured factors, not verified explanations or adjusted estimates.

4. Which design distinction is correct?

D. Random assignment allocates conditions; random sampling selects sample members.

The two random processes concern different stages and do not substitute for one another.

01 Conversation answers

Research Questions and Study Design

E Dialogue gaps

Pages 10-11

- 1. research question.** Turn 2. The research question must specify what is being measured and for whom rather than relying on the broad word works.
- 2. outcome variable.** Turn 4. The outcome variable is course completion, not every possible academic benefit that tutoring might produce.
- 3. denominator.** Turn 6. The denominator is forty tutoring participants; pairing it with thirty-two completers yields the eighty-percent completion rate.
- 4. percentage-point difference.** Turn 8. The percentage-point difference subtracts sixty percent from eighty percent, producing twenty points rather than a twenty-percent relative change.
- 5. self-selection.** Turn 10. Self-selection means students chose tutoring, so the groups may differ in ways other than the tutoring exposure.
- 6. association.** Turn 12. Association describes the observed relationship without claiming that the current design isolates a causal effect.
- 7. Random assignment.** Turn 14. Random assignment allocates study conditions; selecting additional records randomly would not change the original self-selected tutoring exposure.
- 8. external validity.** Turn 16. External validity concerns application beyond the studied course and cohort, which the supplied records do not establish.
- 9. causal effect.** Turn 18. A causal effect is the contribution attributable to tutoring, which requires a suitable design beyond the current descriptive comparison.
- 10. study protocol.** Turn 20. The study protocol should record the precise question, definitions, and analysis boundaries so later reporting follows the same plan.

F Transfer conversation

Page 12

1. 60

Researcher: "The attendee completion rate is 60 percent."

Eighteen completers divided by thirty attendees equals sixty percent.

2. 50

Supervisor: "The nonattendee rate is 50 percent."

Twenty completers divided by forty nonattendees equals fifty percent.

3. 10

Researcher: "The difference is 10 percentage points."

Sixty percent minus fifty percent is ten percentage points.

4. causal effect

Supervisor: "The contrast establishes an association, not a causal effect."

Self-selection and unmeasured differences prevent this comparison from isolating causation.

02 Briefing and language checks

Grant Proposals and Specific Aims

A Briefing checks / page 13

1. What is the pilot designed to assess?

B. Recruitment and follow-up feasibility

The brief defines the pilot's scope as procedures, not efficacy or universal conclusions.

2. What was the observed retention rate?

B. 80%

Twenty-four retained students out of thirty recruited equals eighty percent.

3. Did both local progression criteria pass?

A. No; recruitment passed and retention did not.

Recruitment met its threshold, but eighty-percent retention fell below the specified eighty-five percent.

D Language checks / page 18

1. Which specific aim matches the supplied pilot?

B. Assess recruitment and follow-up procedures to refine a later study.

Recruitment and follow-up procedures are the pilot's actual focus; retention is not an attainment measure, and both stated criteria still apply.

2. How many retained participants would meet at least 85% of thirty?

D. Twenty-six

Twenty-six out of thirty exceeds eighty-five percent, whereas twenty-five is below it.

3. What is the correct progression statement?

C. Recruitment passed, but retention requires redesign or explicit review before progression.

Both criteria were required, and the retention shortfall prevents automatic progression under that rule.

4. What does the retention result say about efficacy?

C. It does not establish whether the workshop improves attainment.

Retention describes follow-up feasibility, not the workshop's effect on academic attainment.

02 Conversation answers

Grant Proposals and Specific Aims

E Dialogue gaps

Pages 19-20

- 1. specific aim.** Turn 2. The specific aim must describe the work this study can address, which is procedural feasibility rather than definitive attainment improvement.
- 2. deliverable.** Turn 4. The deliverable is the practical output of the pilot, here a refined and tested procedure for later research.
- 3. recruitment.** Turn 6. Recruitment concerns bringing students into the study; thirty within four weeks meets the supplied local recruitment threshold.
- 4. retention.** Turn 8. Retention is twenty-four of thirty recruited students, or eighty percent, at the specified follow-up point.
- 5. progression criterion.** Turn 10. The progression criterion requires at least eighty-five percent, which with thirty recruited students needs at least twenty-six retained.
- 6. contingency.** Turn 12. A contingency addresses the failed milestone through redesign or explicit review rather than ignoring or silently changing the predefined rule.
- 7. Feasibility.** Turn 14. Feasibility concerns whether the proposed methods can be carried out, which is distinct from the intervention's effect on attainment.
- 8. funding decision.** Turn 16. A funding decision is made by the relevant funder and is not guaranteed by satisfying an internal planning milestone.
- 9. scope.** Turn 18. Scope defines what the pilot will do and claim, preventing the long-term research ambition from becoming an unsupported present promise.
- 10. rationale.** Turn 20. The rationale explains why the bounded pilot work remains valuable while acknowledging the evidence and unresolved milestone.

F Transfer conversation

Page 21

1. met

Researcher: "The recruitment condition was met."

Twenty students were recruited within the stated three-week window.

2. 85

Adviser: "Retention was 85 percent."

Seventeen divided by twenty equals eighty-five percent, below ninety percent.

3. 18

Researcher: "At least 18 retained students were required."

Eighteen out of twenty equals the required ninety-percent retention threshold.

4. not supported

Adviser: "Automatic progression is not supported under the stated rule."

Only recruitment passed, while the rule requires both conditions to be satisfied.

03 Briefing and language checks

Lab Meetings and Data Challenges

A Briefing checks / page 22

1. What changed between the runs?

D. Exposure time increased from ten to twenty seconds.

Instrument files establish different exposure times for the two recorded measurements.

2. What is not yet established?

C. Whether exposure time explains the entire signal difference

The method difference is known, but its full contribution to the result has not been tested.

3. How should the documentation be corrected?

D. Add a dated correction while preserving the original record.

Transparent correction preserves provenance without concealing the original omission or inventing a misconduct finding.

D Language checks / page 27

1. Which sentence accurately reports the signal contrast?

C. Run B is 60 units, or 50%, higher than Run A.

One hundred eighty minus one hundred twenty is sixty, which is half of the original value.

2. Why is dividing Run B by two not justified as a correction?

C. Proportional signal response to exposure has not been established.

Scaling assumes a relationship that the brief explicitly says has not been verified.

3. What is the next useful comparison?

B. A documented repeat with matched settings and preserved raw records

Matching and documenting conditions helps examine the discrepancy without concealing existing evidence.

4. Which conclusion should be avoided?

C. The omitted note proves research misconduct.

An omission requires examination and correction but does not by itself establish a misconduct finding.

03 Conversation answers

Lab Meetings and Data Challenges

03

E Dialogue gaps

Pages 28-29

- 1. instrument setting.** Turn 2. The instrument setting difference is established by the files: exposure was ten seconds in one run and twenty in the other.
- 2. raw data.** Turn 4. Raw data preserves the original observations and settings, allowing later checks without losing an inconvenient result.
- 3. linear response.** Turn 6. A linear response can include an offset; proportional rescaling also requires a justified zero or background correction and a validated range.
- 4. saturation.** Turn 8. Saturation could make the signal response nonproportional, but it is only a possibility to investigate in this case.
- 5. dated correction.** Turn 10. A dated correction records the discovery transparently while preserving the original note and the history of the omission.
- 6. provenance.** Turn 12. Provenance records the origin and history of the data and corrections, making the discrepancy traceable without prejudging responsibility.
- 7. controlled comparison.** Turn 14. A controlled comparison holds relevant settings consistent so the team can investigate the discrepancy more meaningfully.
- 8. technical replicates.** Turn 16. The planned matched measurements assess measurement variability on the same material, not additional independent biological samples.
- 9. hypothesis.** Turn 18. The hypothesis proposes that the method change contributes to the discrepancy, but it still requires testing rather than being reported as established.
- 10. repeatability check.** Turn 20. The same operator, instrument, method, location, and session specify repeatability conditions, unlike a changed-condition measurement reproducibility study.

F Transfer conversation

Page 30

1. difference

Researcher: "The gain change is a verified method difference."

The instrument records establish a setting difference, not merely a suspected change.

2. untested

Lead: "Its causal contribution remains untested."

No check has isolated the effect of the setting change.

3. dated correction

Researcher: "The notebook needs a dated correction."

A dated correction transparently records the omission without rewriting the past.

4. preserved

Lead: "Both original files should be preserved."

Retaining original records allows later verification and avoids selective removal of evidence.

04 Briefing and language checks

Research Ethics and Human Subjects

A Briefing checks / page 31

1. Why are the coded interviews not automatically anonymous to this team?

C. The team can access the code-to-participant key.

The accessible linking key allows the team to connect codes with participants.

2. Who determines the permitted route under the supplied local rule?

C. The designated ethics office

The fictional institutional procedure assigns the determination to its designated ethics office.

3. What can Tessa do now?

D. Prepare the proposal and gather the existing approval and consent documents.

Preparatory documentation is permitted, while the new use and sharing must await the required determination.

D Language checks / page 36

1. Which description should be sent to the ethics office?

C. Coded interviews with a linking key accessible to the research team

The correct description accurately reports the team's actual ability to link codes with people.

2. What does the old approval establish about this new purpose?

D. Nothing conclusive until its scope is checked against the proposal.

The original materials have not been reviewed against the new use, so their coverage cannot be assumed.

3. Which message to a collaborator is accurate?

D. "The proposed access awaits the designated ethics determination."

The correct message states the pending status without inventing approval or participant agreement.

4. Which preparatory step respects the supplied boundary?

A. Gather documents and describe the proposed use without beginning analysis or sharing.

The local procedure permits proposal preparation while holding the proposed research use and disclosure.

04 Conversation answers

Research Ethics and Human Subjects

E Dialogue gaps

Pages 37-38

1. review status. Turn 2. The review status is unresolved: the required determination has not been made, so the new analysis cannot start under the supplied rule.

2. coded data. Turn 4. Coded data uses substitute labels but may remain identifiable through a linking key, which this team can access.

3. linking key. Turn 6. The linking key connects the interview codes to participant identities and must be disclosed in the description of access conditions.

4. consent scope. Turn 8. Consent scope concerns the uses covered by the relevant participant materials, which have not been checked against the new purpose.

5. purpose change. Turn 10. The purpose change distinguishes the proposed employment-discrimination question from the original student-stress research and requires assessment.

6. authorized determination. Turn 12. The authorized determination is made by the designated ethics office, not inferred by collaborators or the researcher from possession of the files.

7. protocol amendment. Turn 14. A protocol amendment is a proposed change to approved work and must not be described as already authorized when none has been granted.

8. access restriction. Turn 16. The access restriction remains in force for the interviews while the determination is pending; a general description does not authorize disclosure.

9. data-use agreement. Turn 18. A data-use agreement can describe intended access and use conditions, but it does not replace the required ethics determination.

10. secondary use. Turn 20. Secondary use is the proposed reuse for a new research purpose, which remains pending rather than already permitted.

F Transfer conversation

Page 39

1. anonymous

Researcher: "The files are coded, not automatically anonymous."

The accessible linking key means coding does not establish anonymity to the team.

2. determination

Coordinator: "The proposed use still awaits an authorized determination."

Sending the request does not complete the required institutional decision.

3. key

Researcher: "Our team can access the linking key."

The brief confirms that the team can link participant codes to identities.

4. pending

Coordinator: "We can accurately report that review is pending."

No decision has been issued, so pending is the supported status.

05 Briefing and language checks

Authorship, Collaboration, and Credit

A Briefing checks / page 40

1. What substantive contribution has Harper made?

C. Designed and performed the main analysis

The brief records Harper's analysis design and execution as the substantive contribution.

2. Does funding alone establish Nolan's authorship under the stated criteria?

B. No

The adopted criteria require more than obtaining funding alone.

3. What should happen before the author list is finalized?

C. Give Harper a genuine opportunity to review, approve, and accept the relevant responsibilities.

A substantive contributor should not be excluded by denying opportunities to fulfill the remaining criteria.

D Language checks / page 45

1. Which statement about Harper is supported?

A. Harper has a substantive analysis contribution and outstanding authorship responsibilities.

The analysis is documented, while review, final approval, and related responsibilities remain to be completed.

2. Which action would unfairly block completion of the criteria?

D. Withholding the manuscript and then citing lack of review as a reason for exclusion

The contributor cannot fulfill review responsibilities if the group denies the opportunity to do so.

3. What does Nolan's funding-only role establish?

B. A contribution to record, but not sufficient authorship under the adopted criteria

Funding is a real contribution but does not alone meet the journal's stated authorship requirements.

4. What remains unresolved after identifying a substantive contribution?

D. Completion of the remaining responsibilities and agreement on the author list and order

Substantive contribution is only part of the process, and the brief leaves final authorship decisions unsettled.

05 Conversation answers

Authorship, Collaboration, and Credit

E Dialogue gaps

Pages 46-47

- 1. contribution statement.** Turn 2. A contribution statement records the actual work performed, providing a factual basis for the credit discussion.
- 2. critical review.** Turn 4. Critical review is an intellectual examination of the manuscript, which Harper should be able to undertake rather than being excluded for not receiving it.
- 3. final approval.** Turn 6. Final approval concerns the identified publication version and cannot be assumed before Harper has seen and agreed to it.
- 4. Accountability.** Turn 8. Accountability connects authorship with responsibility for addressing accuracy and integrity questions, not merely receiving credit.
- 5. funding acquisition.** Turn 10. Funding acquisition is a contribution to recognize, but by itself does not meet the adopted journal's authorship criteria.
- 6. author order.** Turn 12. Author order is a separate collective decision and is not automatically determined by identifying one substantive contribution.
- 7. contributorship.** Turn 14. Contributorship describes each person's work and helps the group avoid overlooking relevant contributions while assessing the author list.
- 8. corresponding author.** Turn 16. The corresponding author handles journal communications, a role distinct from a guaranteed first position in the byline.
- 9. dispute resolution.** Turn 18. Dispute resolution provides a process for unresolved authorship disagreement rather than treating submission or hierarchy as a substitute for agreement.
- 10. byline.** Turn 20. The byline is the eventual list of authors, which should reflect resolved eligibility and agreed responsibilities rather than the current informal draft.

F Transfer conversation

Page 48

1. contribution

Lead: "The methods work is a substantive contribution."

The supplied methods work satisfies the stated substantive-contribution element.

2. critical review

Collaborator: "Receiving the manuscript gives me an opportunity for critical review."

Circulation enables review but does not mean that intellectual review has already occurred.

3. outstanding

Lead: "Final approval remains outstanding."

The brief states that no final approval has yet been recorded.

4. author order

Collaborator: "Eligibility alone does not settle author order."

The ordering decision remains separate and explicitly unresolved in this case.

05

06 Briefing and language checks

Peer Review and Revision Responses

A Briefing checks / page 49

1. What did the study directly measure?

C. Self-reported confidence after one workshop

The measured outcome was post-workshop confidence, not demonstrated learning or change.

2. Which original claim goes beyond the evidence?

C. The workshop improves learning for all adults.

The original claim substitutes learning for confidence and extends beyond the study population and design.

3. What has been done for this revision?

D. Wording is being corrected; no new data or reanalysis has occurred.

The brief permits a reporting correction and explicitly states no new data or reanalysis.

D Language checks / page 54

1. Which revised claim is supported?

A. Sixty volunteer students at one campus reported mean post-workshop confidence of 7.1 out of ten.

The supported statement reports the measured outcome and sample without inventing a baseline or causal design.

2. Which response accurately describes the work?

A. The abstract and conclusion are narrowed; no reanalysis was performed.

The revision changes interpretation and reporting, not the dataset or analysis.

3. What does acknowledging the limitation accomplish?

B. Makes the interpretation more accurate without removing the design constraint

Explicit reporting improves transparency but cannot alter how the completed study was designed.

4. Which locator should the response use?

C. Page and line numbers verified against the revised manuscript

Accurate locations let the reviewer find the actual correction in the submitted revision.

06 Conversation answers

Peer Review and Revision Responses

E Dialogue gaps

Pages 55-56

1. construct. Turn 2. The measured construct was self-reported confidence, not demonstrated learning, so the original claim described a different outcome.

2. post-only measure. Turn 4. A post-only measure records the state after the workshop without a baseline from which to calculate improvement.

3. limitation. Turn 6. The limitation restricts interpretation; making it explicit does not retroactively add a baseline or comparison group.

4. overgeneralization. Turn 8. Overgeneralization extends the conclusion beyond the volunteer, single-campus sample without supporting evidence.

5. revised manuscript. Turn 10. The revised manuscript must use consistent claim boundaries across sections, rather than correcting only one quoted sentence.

6. point-by-point response. Turn 12. A point-by-point response links the specific reviewer concern to the corresponding explanation and revision.

7. page-and-line reference. Turn 14. A page-and-line reference must match the revised version so the reviewer can locate the actual correction.

8. reanalysis. Turn 16. Reanalysis means new analysis of existing data, which the brief explicitly says has not occurred.

9. revision. Turn 18. The revision clarifies the interpretation and limitation rather than eliminating the underlying design constraint.

10. editorial decision. Turn 20. The editorial decision remains with the journal; a valid correction does not guarantee acceptance.

F Transfer conversation

Page 57

1. satisfaction

Author: "The measured outcome was satisfaction."

The survey asks about satisfaction, not directly observed performance.

2. baseline

Reviewer: "The study cannot calculate improvement without a baseline."

A post-only observation lacks the starting measure needed for a before-and-after change.

3. revision

Author: "The work completed here is a wording revision."

The authors changed the claim without collecting data or performing a new analysis.

4. eliminated

Reviewer: "The design limitation is explicit, not eliminated."

More accurate wording does not change the completed study's design.

07 Briefing and language checks

Data Management and Reproducibility

A Briefing checks / page 58

1. What does -999 represent in this dataset?

B. A failed reading coded as missing

The collector explicitly confirms that minus nine hundred ninety-nine is the failed-reading code.

2. What is the mean of the four valid temperatures?

B. 23 degrees Celsius

The valid values sum to ninety-two, and ninety-two divided by four equals twenty-three.

3. How should 04/05/26 be represented unambiguously after confirmation?

C. 2026-05-04

The collector confirms May fourth, twenty twenty-six, which matches the year-month-day representation.

D Language checks / page 63

1. Which data-handling description is correct?

D. Preserve the raw file and convert the sentinel to missing in a documented analysis copy.

Preserving the raw file retains provenance. The documented analysis copy applies the collector's confirmed missing-value rule without inventing a measurement.

2. What does using four valid observations imply?

D. No value has been imputed for the failed reading.

The calculation uses observed values only and does not estimate the failed measurement.

3. Why is the collector's date confirmation necessary?

A. The original numeric label could otherwise be interpreted in different date orders.

The source label is ambiguous without verified knowledge of its intended date convention.

4. Which release package best supports reproducibility?

B. Data, dictionary, processing steps, and the identified software environment

Interpretable data and traceable computation allow another analyst to understand and rerun the result.

07 Conversation answers

Data Management and Reproducibility

E Dialogue gaps

Pages 64-65

1. unit of measurement. Turn 2. The unit of measurement must come from verified collection information; the collector confirms degrees Celsius in this case.

2. missing-value code. Turn 4. The missing-value code represents an absent valid reading, not an observed temperature or a zero measurement.

3. valid observations. Turn 6. The valid observations are twenty, twenty-two, twenty-four, and twenty-six, excluding the failed-reading code from the stated mean.

4. imputation. Turn 8. Imputation would estimate a replacement value, whereas this case calculates a mean from four observed temperatures without filling the missing reading.

5. raw file. Turn 10. The raw file retains the original record, including its sentinel, while documented transformations belong in a separate derived copy.

6. date format. Turn 12. The date format should reflect the confirmed May fourth date, written as 2026-05-04 rather than guessed from regional convention.

7. time zone. Turn 14. The time zone anchors the collection time; 09:00 UTC is more precise than an unspecified nine o'clock.

8. transformation log. Turn 16. The transformation log records conversion of the sentinel and the calculation rule so the result can be traced to the original data.

9. software environment. Turn 18. The software environment identifies the tools and versions used to produce the result, supporting a repeatable computation.

10. computational reproducibility. Turn 20. Computational reproducibility checks whether the specified data and steps reproduce the result, not whether every scientific assumption is thereby validated.

F Transfer conversation

Page 66

1. three

Analyst: "There are three valid observations."

Ten, twelve, and fourteen are valid; the sentinel marks the missing reading.

2. 12

Steward: "Their mean is 12 degrees Celsius."

The three valid values sum to thirty-six, which divided by three equals twelve.

3. original file

Analyst: "The raw sentinel is retained in the original file."

The source record remains unchanged while a derived copy handles missing values.

4. imputed

Steward: "No missing temperature has been imputed."

The calculation uses observed values rather than estimating a replacement temperature.

08 Briefing and language checks

Academic Presentations and Conferences

A Briefing checks / page 67

1. What change was observed in mean response time?

D. A reduction of 60 milliseconds, or 10%

Six hundred minus five hundred forty is sixty milliseconds, one tenth of the initial mean.

2. Which alternative explanations are not isolated?

B. Practice and order effects

The fixed sequence and absent untrained group leave practice and order effects unresolved.

3. What should Noor say about a confidence interval?

B. None is supplied here, so no interval should be invented.

The brief provides no interval calculation, so the response must not fabricate one.

D Language checks / page 72

1. Which opening best answers the audience's question?

C. "No; the study reports a task-specific change, not proof of the general theory."

The correct opening directly rejects the overstatement while preserving the actual observed result.

2. Which causal claim is unsupported?

B. Training alone caused the entire reduction.

The missing comparison group and fixed order leave alternatives to a training-only explanation.

3. Which next step best addresses the limitation?

C. A controlled follow-up designed to separate training from practice and order effects

A stronger comparison design can test competing explanations that the current study does not isolate.

4. Which statement about the mechanism is accurate?

D. The observed response-time change does not by itself establish the memory mechanism.

Task performance and its explanatory mechanism are different claims requiring appropriate evidence.

08 Conversation answers

Academic Presentations and Conferences

E Dialogue gaps

Pages 73-74

1. central finding. Turn 2. The central finding is the observed response-time reduction in the supplied task and sample, not universal proof of the theory.

2. response time. Turn 4. Response time is the measured task duration; sixty milliseconds is ten percent of the initial six-hundred-millisecond mean.

3. within-participant comparison. Turn 6. A within-participant comparison uses observations from the same people, here before and after in a fixed sequence.

4. practice effect. Turn 8. A practice effect is an alternative explanation arising from repeated task exposure, which the current design does not separate from training.

5. mechanism. Turn 10. The mechanism is an explanation for performance, not something established merely by observing lower response times.

6. confidence interval. Turn 12. No confidence interval is supplied for this exercise, so the presenter must not invent an interval or claim certainty.

7. controlled follow-up. Turn 14. A controlled follow-up should address practice and order alternatives rather than merely repeat the original uncontrolled sequence.

8. Generalizability. Turn 16. Generalizability concerns application beyond the observed volunteers and laboratory task, which requires additional relevant evidence.

9. take-home message. Turn 18. The take-home message summarizes the observation and its limits rather than converting a task-specific result into a universal causal claim.

10. unresolved question. Turn 20. The unresolved question is what produced the change and how broadly it applies, not whether the reported means differ arithmetically.

F Transfer conversation

Page 75

1. 80

Presenter: "The mean reduction is 80 milliseconds."

Eight hundred minus seven hundred twenty equals eighty milliseconds.

2. 10

Questioner: "Relative to the initial mean, that is 10 percent."

Eighty divided by eight hundred equals ten percent.

3. practice and order effects

Presenter: "Training has not been isolated from practice and order effects."

The fixed sequence and absent comparison group leave those alternative explanations unresolved.

4. invented

Questioner: "A significance result must not be invented."

No significance test result is supplied in the case.

09 Conversation answers

The smallest p-value was not the primary outcome

E Extended conversation

Pages 76-77

1. primary outcome. Turn 2. The primary outcome was specified before data collection; selecting another outcome after viewing results does not make it the original primary test.

2. exploratory. Turn 6. Exploratory identifies the analysis performed after inspecting the data, rather than presenting it as the prespecified confirmatory test.

3. multiple testing. Turn 10. Multiple testing concerns the twenty outcomes examined; reporting only the smallest unadjusted p-value conceals the selection process relevant to interpretation.

4. p-value. Turn 12. A p-value concerns data under a specified null model and assumptions; subtracting it from one does not give the probability that a hypothesis is true.

5. independent confirmation. Turn 16. Independent confirmation requires a new appropriately planned test, rather than presenting the same inspected data as a fresh confirmatory sample.

6. deviation statement. Turn 20. The deviation statement records departures from the registered plan, including their timing and reasons, without rewriting the original plan after results are known.

F Transfer exchange

Page 78

1. threshold

Researcher: "The registered test did not meet the specified threshold."

The primary p-value of 0.12 is above the stated 0.05 threshold.

2. exploratory

Collaborator: "The twelve later tests are exploratory."

The tests were not in the registered plan and followed data inspection.

3. unadjusted

Researcher: "The reported minimum p-value is unadjusted."

The brief explicitly states that the selected 0.02 value has no multiple-testing adjustment.

4. confirmation

Collaborator: "Reusing these inspected data is not independent confirmation."

The same inspected dataset is not a new independent test of the selected result.

09

Background reading: [Center for Open Science: preregistration](#). The conversation is an original fictional teaching case.

10 Conversation answers

The indirect-cost rate applies to the base, not every line

E Extended conversation

Pages 79-80

1. cost base. Turn 2. The cost base is the amount to which the rate applies; the supplied foundation terms exclude equipment from that amount.

2. indirect costs. Turn 6. Indirect costs are \$5,200 under the supplied 10% rate and \$52,000 base, not 10% of every direct line.

3. budget justification. Turn 10. The budget justification must connect requested costs to the project and permitted categories; room below a ceiling does not itself justify a new charge.

4. personnel total. Turn 12. The personnel total already includes benefits in this case; applying another benefits charge would double-count the same supplied cost.

5. cost sharing. Turn 16. Cost sharing means project costs supported from another source; the \$800 unused ceiling is not an institutional contribution or expenditure.

6. total request. Turn 20. The total request combines \$64,000 direct costs with \$5,200 indirect costs, for \$69,200 subject to submission approval and the funder's decision.

F Transfer exchange

Page 81

1. 40,000

Investigator: "The indirect-cost base is 40,000 dollars."

Subtracting the ten-thousand equipment exclusion from fifty thousand leaves forty thousand.

2. 6,000

Officer: "The indirect-cost amount is 6,000 dollars."

Fifteen percent of the forty-thousand eligible base equals six thousand.

3. 56,000

Investigator: "The combined request is 56,000 dollars."

All fifty thousand direct costs remain in the request, plus six thousand indirect costs.

4. 4,000

Officer: "The unused ceiling is 4,000 dollars, not an award."

The sixty-thousand ceiling exceeds the fifty-six-thousand request by four thousand dollars.

Background reading: [NIH: developing a budget; check the actual award terms](#). The conversation is an original fictional teaching case.

11 Conversation answers

Which manuscript can the repository release?

E Extended conversation

Pages 82-83

1. accepted manuscript. Turn 2. The accepted manuscript includes the peer-reviewed changes accepted for publication but is distinct from the publisher's formatted PDF.

2. version of record. Turn 6. The version of record is the formally published article; permission to link to it does not grant permission to deposit its PDF.

3. embargo. Turn 10. The embargo delays public access to the deposited manuscript until the stated date; it does not prevent the permitted closed deposit in this case.

4. third-party material. Turn 12. Third-party material can have separate permissions; the article's deposit permission does not automatically resolve the rights to the reproduced chart.

5. metadata. Turn 16. Metadata describe and identify the article and its version; they help users find the published work without substituting for full-text permissions.

6. release conditions. Turn 20. The release conditions include the October 15 date and resolution of the chart permissions; deposit alone does not satisfy both conditions.

F Transfer exchange

Page 84

1. manuscript

Researcher: "The permitted file is the accepted manuscript."

The agreement permits the accepted manuscript rather than the excluded publisher PDF.

2. embargo

Librarian: "Public full-text access remains closed during the embargo."

The public-release date has not arrived, although deposit is permitted now.

3. December 1

Researcher: "The earliest stated public-release date is December 1."

The supplied agreement names December first, not the current November date.

4. excluded

Librarian: "The publisher PDF remains excluded."

The date condition does not expand permission to a file version the agreement excludes.

11

Background reading: [NISO: Journal Article Versions](#). The conversation is an original fictional teaching case.

Find the move you need

Use these as precise language models. Replace details only when you know the actual facts, authority, and agreed conditions.

01 Research Questions and Study Design

Clarify the broad verb: "What outcome would count as working?"

Name the outcome: "We are measuring course completion."

Express the contrast: "That is a 20-percentage-point difference."

Full phrase set: pages 7-8 | Dialogue: pages 10-11

02 Grant Proposals and Specific Aims

Separate ambition from aim: "The long-term goal is broader than this pilot deliverable."

Name the recruitment target: "The local target was 30 students in four weeks."

Apply the threshold: "That is below the predefined 85% criterion."

Full phrase set: pages 16-17 | Dialogue: pages 19-20

03 Lab Meetings and Data Challenges

Open with the result: "Run A gave 120 units; Run B gave 180."

Disclose the omission: "The summary notes did not record that change."

Limit the explanation: "The change may contribute, but we have not isolated its effect."

Full phrase set: pages 25-26 | Dialogue: pages 28-29

04 Research Ethics and Human Subjects

Describe the new question: "The proposed reuse concerns employment discrimination."

Avoid an anonymity claim: "Coding does not make these data anonymous to us."

State the hold: "Do not start the new analysis or sharing yet."

Full phrase set: pages 34-35 | Dialogue: pages 37-38

Keep the conversation productive

Use these as precise language models. Replace details only when you know the actual facts, authority, and agreed conditions.

05 Authorship, Collaboration, and Credit

Open factually: "Let us document what each collaborator contributed."

Identify what remains: "The manuscript review and final approval have not happened yet."

Keep titles out of the test: "Job title is not the criterion."

Full phrase set: pages 43-44 | Dialogue: pages 46-47

06 Peer Review and Revision Responses

Acknowledge the point: "We agree that the original claim exceeded the design."

Keep the observation: "The observed mean remains 7.1 out of ten."

Describe the revision: "We have narrowed the abstract and conclusion."

Full phrase set: pages 52-53 | Dialogue: pages 55-56

07 Data Management and Reproducibility

Ask for meaning: "What does the variable t represent?"

Reject a false zero: "Missing is not the same as zero."

Give the result with units: "The valid-observation mean is 23 degrees Celsius."

Full phrase set: pages 61-62 | Dialogue: pages 64-65

08 Academic Presentations and Conferences

Answer directly: "No, this study does not prove the general theory."

Name the sample: "All 24 volunteers completed the task."

Avoid a causal shortcut: "We have not isolated training from those effects."

Full phrase set: pages 70-71 | Dialogue: pages 73-74

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

abstract revise the abstract	50
access restriction apply access restrictions	33
accountability accept accountability	42
acknowledgment agree an acknowledgment	42
analysis copy create a documented analysis copy	59
approach evaluate the proposed approach	15
approval scope verify the approval scope	32
arbitrary unit report arbitrary units	23
association report an association	5
attrition report participant attrition	14
audience question answer an audience question	68
author order discuss author order	41
authorized determination await an authorized determination	33
authorship establish authorship eligibility	41
authorship criteria apply authorship criteria	42
baseline difference examine baseline differences	6
baseline measure identify the missing baseline measure	51
batch effect investigate a batch effect	23
biological replicate distinguish biological replicates	24
boundary condition identify boundary conditions	69
byline finalize the byline	41
causal effect estimate a causal effect	6
central finding state the central finding	68
claim qualify a claim	50

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

clarifying question ask a clarifying question	69
coded data describe coded data accurately	32
cohort follow a student cohort	5
collaboration agreement revisit the collaboration agreement	42
collection timestamp preserve collection timestamps	60
comparison arm include a comparison arm	68
comparison condition specify a comparison condition	51
comparison group identify the comparison group	5
complete-case calculation describe a complete-case calculation	60
completion rate calculate the completion rate	6
computational reproducibility check computational reproducibility	60
conceptualization describe conceptualization	41
conclusion narrow the conclusion	50
conference presentation deliver a conference presentation	68
confidence interval report a confidence interval	69
confidentiality maintain confidentiality	33
confounder address potential confounders	6
consent scope check the consent scope	32
construct identify the measured construct	51
contingency specify a contingency	15
contribution statement prepare a contribution statement	42
contributorship document contributorship	41
controlled comparison plan a controlled comparison	24
controlled follow-up propose a controlled follow-up	69

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

corresponding author identify the corresponding author	42
counterbalancing plan appropriate counterbalancing	68
critical review contribute critical review	42
data cleaning document data cleaning	59
data curation describe data curation	41
data dictionary publish a data dictionary	59
data validation perform data validation	60
data-use agreement establish a data-use agreement	33
date format standardize the date format	60
dated correction add a dated correction	24
de-identification document de-identification	33
definitive study plan a definitive study	15
deliverable identify the pilot deliverable	15
denominator state the denominator	6
dependency list analysis dependencies	60
dispute resolution use the dispute-resolution process	42
editorial decision await the editorial decision	51
efficacy distinguish efficacy from feasibility	14
estimand define the estimand	6
ethics review request ethics review	32
exemption determination obtain an exemption determination	32
experimental run compare experimental runs	23
exposure define the exposure	5
exposure time document exposure time	23

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

external validity discuss external validity	6
failed reading mark a failed reading	59
feasibility assess feasibility	14
final approval obtain final approval	42
fixed order acknowledge a fixed order	68
follow-up complete follow-up	14
formal analysis credit formal analysis	41
funding acquisition acknowledge funding acquisition	41
funding decision distinguish a funding decision	15
generalizability assess generalizability	69
honorary authorship avoid honorary authorship	42
human-subjects research assess human-subjects research status	32
hypothesis test a methodological hypothesis	24
identifiability assess identifiability	33
imputation distinguish cleaning from imputation	60
independent replication seek independent replication	69
informed consent review informed-consent materials	32
innovation describe the innovation precisely	15
institutional review board consult the institutional review board	32
instrument setting match instrument settings	23
internal validity assess internal validity	6
investigation record investigation work	41
lab meeting present at a lab meeting	23
lab notebook maintain the lab notebook	24

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

learning assessment distinguish a learning assessment	51
limitation state the study limitation	51
linear response verify a linear response	24
linking key restrict access to the linking key	33
long-term goal state the long-term goal	14
manuscript version circulate the final manuscript version	42
measurement variability quantify measurement variability	24
mechanism test a proposed mechanism	69
metadata supply essential metadata	59
method deviation document a method deviation	23
methodology record methodology contributions	41
milestone set a measurable milestone	14
millisecond report milliseconds	68
missing-value code document missing-value codes	59
normalization justify normalization	24
observational study describe an observational study	5
omitted contributor address an omitted contributor	42
operational definition state an operational definition	6
order effect control for order effects	68
outcome variable specify the outcome variable	5
overgeneralization remove an overgeneralization	50
page-and-line reference provide page-and-line references	51
participant information review participant information	33
peer review respond to peer review	50

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

percentage-point difference	report a percentage-point difference	6
pilot study	design a feasibility pilot	14
point-by-point response	provide a point-by-point response	50
post-only measure	interpret a post-only measure	51
practice effect	consider a practice effect	68
precision	discuss estimate precision	15
processed result	trace a processed result	23
progression criterion	apply progression criteria	14
protocol amendment	submit a protocol amendment	32
protocol version	record the protocol version	23
provenance	preserve data provenance	24
purpose change	describe the purpose change	33
random assignment	distinguish random assignment	5
random sampling	distinguish random sampling	5
rationale	strengthen the study rationale	15
raw data	preserve raw data	23
raw file	preserve the raw file	59
readme	include a readme	60
reanalysis	distinguish revision from reanalysis	51
recontact	assess a recontact proposal	33
recruitment	monitor recruitment	14
recruitment window	define the recruitment window	14
reference material	measure the reference material	23
relative difference	distinguish a relative difference	6

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

repeatability check	document a repeatability check	24
repository	select a suitable repository	60
research ethics committee	seek research ethics committee review	32
research question	refine a research question	5
response letter	prepare a response letter	50
response time	measure response time	68
retention	calculate follow-up retention	14
review status	state the review status	33
reviewer comment	address a reviewer comment	50
revised manuscript	check the revised manuscript	50
revision	document the revision	50
sample size	justify the sample size	15
sampling frame	describe the sampling frame	50
saturation	check for signal saturation	24
scope	keep the aims within scope	15
secondary use	propose a secondary use	32
self-report	distinguish self-report from direct measurement	51
self-selection	account for self-selection	5
sentinel value	identify a sentinel value	59
signal intensity	compare signal intensity	24
significance	explain research significance	15
significance test	interpret a significance test	69
single-site study	state the single-site limitation	51
software environment	document the software environment	60

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

specific aim refine a specific aim	14
statistical uncertainty communicate statistical uncertainty	69
study objective align the study objective	15
study protocol revise the study protocol	6
study sample describe the study sample	5
substantive contribution recognize a substantive contribution	41
supervision distinguish supervision from authorship	41
take-home message clarify the take-home message	68
target population define the target population	5
technical replicate compare technical replicates	23
theoretical claim bound a theoretical claim	69
time allocation respect the time allocation	69
time zone record the time zone	60
tracked changes supply tracked changes	50
transformation log maintain a transformation log	59
unit of measurement specify units of measurement	59
unresolved question state the unresolved question	69
valid observation count valid observations	59
variable name clarify variable names	59
version identifier record the version identifier	60
voluntariness protect voluntariness	33
volunteer sample acknowledge the volunteer sample	51
waiver distinguish a requested waiver	32
within-participant comparison describe a within-participant comparison	68

Original cases. Clear boundaries.

These are original fictional teaching conversations, not transcripts of real workers. Names, figures, organizations, and incidents are invented. Sources support terminology and communication aims, not the fictional results or an endorsement of this book.

[1] National Center for Complementary and Integrative Health. Pilot Studies: Common Uses and Misuses.

[Open the original source](#)

Background on feasibility pilots and limits on efficacy claims. The learning-study milestones in this book are original fictional criteria, not a funder requirement. Checked 30 September 2026.

[2] U.S. Office for Human Research Protections. Coded Private Information or Biospecimens Used in Research.

[Open the original source](#)

U.S. guidance on coded information and authorized determinations for secondary research. The exercises supply a fictional local review requirement and do not determine exemption status. Checked 30 September 2026.

[3] International Committee of Medical Journal Editors. Defining the Role of Authors and Contributors.

[Open the original source](#)

Background for the authorship case, which explicitly adopts these criteria. Authorship rules and ordering conventions differ among journals and disciplines. Checked 30 September 2026.

[4] National Institutes of Health. Writing a Data Management and Sharing Plan.

[Open the original source](#)

Background for data documentation, methods, tools, and appropriate sharing. The environmental dataset and laboratory records are original teaching cases. Checked 30 September 2026.

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